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SPLICE SWITCHING OLIGONUCLEOTIDES TARGETING PSEUDOEXONS

Abstract

The present invention relates to a method for identifying splice switching oligonucleotides (SSOs) able to modulate expression of a target protein in a cell by promoting incorporation of a pseudoexon into the mature mRNA upon binding to the pre-mRNA in the region +9 to +39 downstream to the 5' splice site of said pseudoexon. The invention also relates to SSO obtained by said method and uses thereof.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to co-pending PCT International Patent Application Serial No. PCT/EP2022/051790, filed Jan. 26, 2022, which claims priority to EP Patent Application No. 21153508.3, filed on Jan. 26, 2021, the entire content of both of which is incorporated herein by reference.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically as a file in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII format file, created on Nov. 24, 2023, is named Corrected_Sequence_list_16SYDD-PV10202PA.txt and is 34,109 bytes in size.

TECHNICAL FIELD OF THE INVENTION

[0003] The present invention relates to splice switching oligonucleotides (SSOs) that can activate splicing of pseudoexons. In particular, the SSOs are able to promote inclusion of the pseudoexon in an mRNA transcript of a gene, thereby inhibiting expression of a functional gene product. In another aspect, the invention relates to a method for identifying pseudoexons for which it is possible to incorporate the pseudoexons in mature mRNAs using the SSOs.

BACKGROUND OF THE INVENTION

[0004] Newly synthesized eukaryotic mRNA molecules, also known as primary transcripts or pre-mRNA, made in the nucleus, are processed before or during transport to the cytoplasm for translation. Processing of the pre-mRNAs includes addition of a 5' methylated cap and an approximately 200-250 nucleotides poly(A) tail to the 3' end of the transcript.

[0005] Another step in mRNA processing is splicing of the pre-mRNA, which is part of the maturation of 90-95% of mammalian mRNAs. Introns (or intervening sequences) are regions of a primary transcript that are not included in the coding sequence of the mature mRNA. Exons are regions of a primary transcript that remain in the mature mRNA when it reaches the cytoplasm. The exons are spliced together to form the mature mRNA sequence. Splicing occurs between splice sites that together form a splice junction. The splice site at the 5' end of the intron is often called the "5' splice site," or "splice donor site" and the splice site at the 3' end of the intron is called the "3' splice site" or "splice acceptor site". In splicing, the 3' end of an upstream exon is joined to the 5' end of the downstream exon. Thus, the unspliced RNA (or pre-mRNA) has an exon/intron splice site at the 5' end of an intron and an intron/exon splice site at the 3' end of an intron. After the intron is removed, the exons are contiguous at what is sometimes referred to as the exon/exon junction or boundary in the mature mRNA. Alternative splicing, defined as the splicing together of different combinations of exons or exon segments, often results in multiple mature mRNA transcripts expressed from a single gene.

[0006] The splicing of precursor mRNA (pre-mRNA) is an essential step in eukaryotic gene expression, where introns are removed through the activities of the spliceosome, and the coding parts of a gene are spliced together, resulting in a functional mRNA. Pre-mRNA splicing is a highly controlled process, and it is well established that mutations can impact splicing and generate aberrant transcripts. Correct mRNA splicing depends on regulatory sequences, which are

recognized by different factors of the spliceosome, as well as splicing regulatory factors. The splicing regulatory factors either stimulate or repress recognition and splicing of exons by sequence specific binding to splicing regulatory sequences such as splicing enhancers and splicing silencers. Pre-mRNA splicing in eukaryotes is often associated with extensive alternative splicing to enrich their proteome. Alternative selection of splice sites permits eukaryotes to modulate cell type specific gene expression, contributing to their functional diversification. Alternative splicing is a highly regulated process influenced by the splicing regulatory proteins, such as SR proteins or hnRNPs, which recognize splicing regulatory sequences, such as exonic splicing enhancers (ESEs) and exonic splicing silencers (ESSs) in exons, and intronic splicing enhancers (ISEs) and intronic splicing silencers (ISSs) in introns.

[0007] It is a well-known fact that exonic mutations, which either create or eliminate existing splicing regulatory sequences other than the splice site sequences often lead to mis-splicing of the RNA that might result in diseases. However, it is difficult to predict which mutations affect splicing as not all exons are critically dependent on splicing regulatory elements other than the splice sites, and consequently only a limited number of exons are vulnerable to mutations in splicing regulatory sequences outside of the splice sites.

SUMMARY OF THE INVENTION

[0008] The present invention relates to the identification of sequence parameters in a gene comprising a pseudoexon, which can be used to determine if it is possible to get the pseudoexon incorporated into the mature mRNA using a splice switching oligonucleotide (SSO). Incorporation of pseudoexons can e.g. be used to inactivate, disrupt, or alter the function of the functional product expressed from a gene by incorporation of the pseudoexon in the mature mRNA (see also example 1 and corresponding figure and figure legend for further information). The invention also relates to (medical) uses of such SSOs.

[0009] Example 2 shows that the identified parameters are essential for identifying activatable pseudoexons (Table 1) and non-activatable pseudoexons (Table 2).

[0010] Example 3 shows data in relation to SMAD2 (see also example 11).

[0011] Example 12 shows data in relation to RNF115.

[0012] Examples 4-11 and 13-14 show further examples for specific genes comprising pseudoexons where the pseudoexons can be activated (incorporated in the mature mRNA).

[0013] Thus, an object of the present invention relates to the provision of sequence parameters (criteria) which can identify binding sites for SSOs for incorporation of pseudoexons into mature mRNA.

[0014] Another object of the invention is to provide SSOs which, in vivo, can promote incorporation of pseudoexons into mature mRNA, thereby inactivating, disrupting, or altering the natural function of genes.

[0015] Thus, one aspect of the invention relates to a method for identifying SSOs able to modulate expression and/or function of a target protein in a cell by promoting incorporation of a pseudoexon into the mature mRNA upon binding to the pre-mRNA in the region +9 to +39 downstream to the 5' splice site of said pseudoexon, the method comprising; [0016] a) providing one or more gene sequences comprising one or more identified pseudoexons, such as in the form of a database or other storage means; [0017] b) determining for the one or more gene sequences; if the pseudoexon meets the following criteria: [0018] Pseudoexon length <160 nt; [0019] Pseudoexon length >30 nt; [0020] The last 3 nt of the pseudoexon are different from TAG; [0021] Donor splice site has a MaxEnt score ≥ 4.33 ; [0022] Donor splice site has a MaxEnt score ≤ 10.06 ; and [0023] Acceptor splice site has a MaxEnt score ≥ 3.63 ; [0024] c) determining for the one or more gene sequences if the sequence region +9 to +39 downstream to the 5' splice site of said pseudoexon (3) meets the following criteria: [0025] Total pyrimidines ≤ 20 ; [0026] Total thymidine bases ≤ 12 ; [0027] Total thymidine bases ≥ 4 ; [0028] Total guanine bases ≤ 12 ; [0029] Maximum length of thymidine polymer ≤ 4 ; [0030] Maximum length of pyrimidine polymer ≤ 10 ; [0031] Minimum length of

purine polymer ≥ 3 ; and [0032] Maximum number of guanine polymers of at least 3 nt length ≤ 2 ; wherein, if one or more gene sequences meet the criteria according to point b) and point c), said region +9 to +39 downstream to the 5' splice site of said pseudoexon (3) is considered a target for an SSO able to, in vivo, hybridize to the pre-mRNA (2) of said gene within the region +9 to +39 downstream to the 5' splice site of said pseudoexon (3); and resulting in said pseudoexon (3) becoming part of the mature mRNA to a greater extent compared to corresponding pre-mRNA not contacted with the SSO (1);

and

optionally, producing said SSO, optionally for use as a medicament.

[0033] The present invention also relates to specifically identified SSO for use as medicaments.

Thus, another aspect of the invention relates to a composition comprising a splice switching oligonucleotide (SSO) for use as a medicament, said composition comprising [0034] an SSO complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0035] a nucleic acid according to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0036] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0037] a nucleic acid sequence having at least 90% sequence identity to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201;

or [0038] an SSO selected from the group consisting of: [0039] a nucleic acid according to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or [0040] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or [0041] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216;

wherein said SSO being complementary or substantially complementary to a target pre-mRNA (encoding a functional disorder-causing or disorder-influencing protein), said target pre-mRNA comprising: [0042] a function-disabling pseudoexon comprising: [0043] at the 5'-end a 3' splice site; and [0044] at the 3'-end a 5' splice site;

wherein said SSO is complementary or substantially complementary to the target pre-RNA at a region +9 to +39 downstream to the 5' splice site of said pseudoexon;

wherein, when said SSO, in vivo, hybridizes to the pre-mRNA within the region +9 to +39 downstream to the 5' splice site of said pseudoexon; said pseudoexon becomes part of the mature mRNA to a greater extent compared to corresponding pre-mRNA not contacted with the SSO.

[0045] In yet an aspect the invention relates to a [0046] a composition comprising an SSO complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0047] a nucleic acid according to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0048] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0049] a nucleic acid sequence having at least 90% sequence identity to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201;

or [0050] a composition comprising an SSO selected from the group consisting of: [0051] a nucleic acid according to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or [0052] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or [0053] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0054] FIG. 1

[0055] FIG. 1 shows a schematic overview of gene expression regulation by SSO according to the invention. FIG. 1A) a Sweet Spot region is identified in the intron part of the pre-mRNA as a region +9 to +39 nucleotides downstream for the 5'splice site of a pseudoexon that obeys the criteria according to the invention. FIG. 1B) Pseudoexon inclusion into the mRNA transcript can be activated and increased by employing SSOs complementary to the Sweet Spot region of pseudoexons fulfilling the criteria. Pseudoexon inclusion into the mRNA will modulate gene expression either at the mRNA level or protein level, by mislocalization, destabilization, degradation or alteration of mRNA or protein function.

[0056] FIG. 2

[0057] FIG. 2 shows a schematic demonstration of how RNA-sequencing data can be used in detection of in vivo spliced double junctions for empirical detection of pseudoexons, which are included into the endogenous transcript at low levels.

[0058] After mapping to the human genome, reads are filtered to retain only fragments containing at least two splicing junctions. The splicing junctions of the entire fragment are then assembled into an exon structure, allowing for an unmapped gap between reads in the fragment of up to 100 bp. Exons are then classified using known exon annotations to identify pseudoexons contained within introns. Novel pseudoexons that may be candidates for activation by SSOs binding to the Sweet Spot region can be identified by 14 criteria, after which highly therapeutically relevant pseudoexons can be identified in genes where a downregulation of expression or alteration of the functional gene product is medically relevant.

[0059] FIG. 3

[0060] LINGO2 pseudoexon inclusion inhibits growth and proliferation of glioblastoma cells. (A) RT-PCR analysis of LINGO2 pseudoexon splicing in U251 cells transfected with the LINGO2 pseudoexon +11 SSO and a nontargeting SSO control. The upper band includes the pseudoexon, which is activated by transfection of the +11 SSO. (B) IncuCyte® cell proliferation assay showing growth curves of U251 cells transfected with the LINGO2+11 SSO and a nontargeting SSO control at different concentrations (cell confluency relative to time after transfection). The growth is inhibited by transfection of the +11 SSO in a dose-dependent manner.

[0061] FIG. 4

[0062] TAF2 pseudoexon inclusion inhibits growth and proliferation of lung cancer cells. (A) RT-PCR analysis of TAF2 pseudoexon splicing in NCI-H358 cells transfected with the TAF2 pseudoexon +11 SSO and a nontargeting SSO control. The upper band includes the pseudoexon, which is activated by transfection of the +11 SSO. (B) IncuCyte® cell proliferation assay showing growth curves of NCI-H358 cells transfected with the TAF2 pseudoexon +11 SSO and a nontargeting SSO control (cell confluency relative to time after transfection). The growth is inhibited by transfection of the +11 SSO. (C) Bar plots from WST-1 cell viability and growth assay showing absorbance at 450 nm as a measure of cell viability of NCI-H358 lung cancer cells transfected with the TAF2 pseudoexon +11 SSO and a nontargeting SSO control. The cell viability is decreased by transfection of the +11 SSO.

[0063] FIG. 5

[0064] Optimization of TRPM7 SSOs targeting the Sweet Spot region. (A+B) RT-PCR analysis of TRPM7 pseudoexon splicing in (A) HeLa and (B) U251 cells transfected with 20 nM of the TRPM7 +9, +10, +11, +12 and +13 SSOs, including controls; transfection of a nontargeting control SSO and untransfected cells with transfection reagent (RNAiMAX) or without (UTR). The upper band includes the pseudoexon and the amount is increased by transfection of the +9 to +13 SSOs. (C) Bar plots from IncuCyte® cell proliferation assay showing the relative cell count 68 hours after transfection of the TRPM7 pseudoexon +13 SSO and TRPM7 siRNA (KD), including controls; transfection of a nontargeting control SSO and untransfected cells with transfection reagent (RNAiMAX) or without (UTR). The growth is inhibited by transfection of the +13 SSO. Student's t test, * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$.

[0065] FIG. 6

[0066] Optimization of HIF1A SSOs targeting the Sweet Spot region. (A) RT-PCR analysis of HIF1A pseudoexon splicing in U251 cells transfected with the HIF1A pseudoexon +9, +10, +11, +12 and +13 SSOs. The middle band includes the pseudoexon and the efficiency is highest by transfection with the +10 SSO. Pseudoexon inclusion levels were quantified using the Fragment Analyzer (Advanced Analytical Technologies). (B) WST-1 assay showing absorbance as a measure of growth and proliferation of U251 cells transfected with 20 nM of the HIF1A pseudoexon +10 SSO and a nontargeting SSO control at normoxic (N) or hypoxic (H) conditions. Student's t test, * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$. (C) Western blot of protein extracted from PANC-1 cells transfected with 20 nM of the HIF1A pseudoexon +10 SSO, a nontargeting SSO control and untransfected cells (UT) at normoxic or hypoxic conditions, using a HIF-1 α specific antibody and a β -actin specific antibody as loading control. HIF-1 α protein is not present at normoxia, but produced at hypoxic conditions in control. Translation of HIF-1 α protein is reduced by the +10 SSO at hypoxic conditions.

[0067] FIG. 7

[0068] RNF115 pseudoexon inclusion leads to reduced RNF115 proteins levels and inhibits growth of lung adenocarcinoma cells. (A) RT-PCR analysis of RNF115 pseudoexon splicing in NCI-H23 cells transfected with the RNF115 pseudoexon +11 SSO and a nontargeting SSO control. The upper band includes the pseudoexon, which is activated by transfection of the +11 SSO. (B) WST-1 assay showing growth of NCI-H23 cells transfected with the RNF115 +11 SSO and a nontargeting SSO control at different. The growth is inhibited by transfection of the +11 SSO in a dose-dependent manner. (C) Western blot of protein extracted from NCI-H23 cells transfected with 5, 10 and 20 nM of the RNF115 pseudoexon +11 SSO, a nontargeting SSO control and untransfected cells (UT), using a RNF115 specific antibody, a β -catenin antibody and a β -actin specific antibody as loading control. Protein levels of RNF115 and β -catenin are reduced by the +11 SSO.

[0069] FIG. 8

[0070] SMAD2 pseudoexon inclusion decreases fibrosis in hepatic stellate cells. (A) RT-PCR analysis of SMAD2 pseudoexon splicing in HeLa cells transfected with the SMAD2 +11 SSO and a nontargeting SSO control. The upper band includes the pseudoexon which is activated by transfection with the +11 SSO. (B) RT-PCR analysis of SMAD2 pseudoexon splicing in LX-2 hepatic stellate cells transfected with the SMAD2 +11 SSO and a nontargeting SSO control. (C) Western blotting analysis of protein from HepG2 liver cells transfected with the SMAD2 SSO, a nontargeting control SSO or untransfected (UTR) HepG2 cells, stimulated with (+) or without (–) TGF β 16 hours before protein harvest. Transfection with the +11 SSO reduces SMAD2, as well as phosphoSMAD2 during TGF β stimulation. (D) LX-2 hepatic stellate cells were transfected with either the SMAD2 +11 SSO or a nontargeting SSO control and stimulated with TGF β for 72 hours. Phase-contrast images were captured of the cells, and the number of differentiating cells were counted in ImageJ. Reduction of SMAD2 with the +11 SSO, decreases myofibroblast formation during TGF β stimulation of fibrosis.

[0071] FIG. 9

[0072] SSO-mediated LRRK2 pseudoexon inclusion. (A) Schematic representation of the consequences of induced LRRK2 pseudoexon inclusion. Inclusion of a 54 nt pseudoexon from LRRK2 intron 47 will introduce 18 amino acids to the WD40 domain of the translated LRRK2 protein, and inclusion of a 82 nt pseudoexon with the same 5' splice site will cause a frame-shift and insertion of a premature termination codon which is a target for transcript degradation by nonsense-mediated mRNA decay (NMD). (B) RT-PCR analysis of LRRK2 pseudoexon splicing in HeLa and U251 cells transfected with the LRRK2 pseudoexon +11 SSO and a nontargeting SSO control (ctrl SSO). The upper bands include the pseudoexons, which is activated for inclusion in mRNA by transfection of the +11 SSO. UT; untransfected.

[0073] The present invention will now be described in more detail in the following.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0074] Prior to discussing the present invention in further details, the following terms and conventions will first be defined:

Pseudoexon (PE)

[0075] In the present context, the terms “pseudoexon” or “PE” relate to exonic-like sequences that are present within intronic regions but are normally ignored by the spliceosomal machinery. Thus, pseudoexons do not, under normal splicing conditions, become part of the mature mRNA or only become part of the mature mRNA at low levels. Thus, pseudoexons are intronic sequences flanked by 3' and 5' splice sites, but pseudoexons are often not annotated due to the normally low inclusion into the mRNA transcript. Moreover, when included, pseudoexons will either disrupt or significantly alter the function of the normal transcript or protein.

Function-Disabling Pseudoexon

[0076] In the present context, the term “function-disabling pseudoexon”, relates to the situation that the presence of the pseudoexon in the mature mRNA results in inactivation, reduced activity, reduced transcription and/or altered function of the protein expressed from the mRNA (compared to mature mRNA without the pseudoexon).

Exonic Splicing Enhancer (ESE)

[0077] As used herein, the terms “Exonic Splicing Enhancer” or “Exon Splicing Enhancer” or “ESE” means a nucleotide sequence, which when present in the exon and accessible for binding of nuclear splicing regulatory proteins and/or by forming a secondary structure or a part thereof of the pre-mRNA stimulates inclusion of this exon into the final spliced mRNA during pre-mRNA splicing.

Exonic Splicing Silencer (ESS)

[0078] As used herein, the terms “Exonic Splicing Silencer” or “Exon Splicing Silencer” or “ESS” mean a nucleotide sequence, which when present in the exon and accessible for binding of nuclear splicing regulatory proteins and/or by forming a secondary structure or a part thereof of the pre-mRNA inhibits inclusion of this exon into the final spliced mRNA during pre-mRNA splicing.

Intronic Splicing Enhancer (ISE)

[0079] As used herein, the terms “Intronic Splicing Enhancer” or “Intron Splicing Enhancer” or “ISE” mean a nucleotide sequence, which when present in the intron and accessible for binding of nuclear splicing regulatory proteins and/or by forming a secondary structure or a part thereof of the pre-mRNA stimulates inclusion of an exon into the final spliced mRNA during pre-mRNA splicing.

Intronic Splicing Silencer (ISS)

[0080] As used herein, the terms “Intronic Splicing Silencer” or “Intron Splicing Silencer” or “ISS” mean a nucleotide sequence, which when present in the intron and accessible for binding of nuclear splicing regulatory proteins and/or by forming a secondary structure or a part thereof of the pre-mRNA inhibits inclusion of an exon into the final spliced mRNA during pre-mRNA splicing.

Splice Sites

[0081] Splice sites at the 5' end of the intron are often called the “5' splice site,” or “splice donor site” and the splice site at the 3' end of the intron are often called the “3' splice site” or “splice acceptor site”.

Nonsense-Mediated mRNA Decay (NMD)

[0082] “Nonsense-mediated mRNA decay” or “NMD” is a surveillance pathway that exists in all eukaryotes. Its main function is to reduce errors in gene expression by eliminating mRNA transcripts that contain premature termination codons (PTCs). In relation to the present invention, the introduction of the pseudoexon may induce NMD if a PTC is present in the introduced pseudoexon or when the pseudoexon changes the reading frame of the mature transcript.

Nucleotide

[0083] As used herein, “nucleotide” means a nucleoside further comprising a phosphate linking

group. As used herein, “linked nucleosides” may or may not be linked by phosphate linkages and thus includes, but is not limited to “linked nucleotides”. As used herein, “linked nucleosides” are nucleosides that are connected in a continuous sequence (i.e., no additional nucleosides are present between those that are linked).

Nucleobase

[0084] As used herein, “nucleobase” means a group of atoms that can be linked to a sugar moiety to create a nucleoside that is capable of incorporation into an oligonucleotide, and wherein the group of atoms is capable of bonding with a complementary naturally occurring nucleobase of another oligonucleotide or nucleic acid. Nucleobases may be naturally occurring or may be modified.

Unmodified Nucleobase

[0085] As used herein the terms, “unmodified nucleobase” or “naturally occurring nucleobase” mean the naturally occurring heterocyclic nucleobases of RNA or DNA: the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) (including 5-methyl C), and uracil (U).

Modified Nucleobase

[0086] As used herein, “modified nucleobase” means any nucleobase that is not a naturally occurring nucleobase.

Modified Nucleoside

[0087] As used herein, “modified nucleoside” means a nucleoside comprising at least one chemical modification compared to naturally occurring RNA or DNA nucleosides. Modified nucleosides comprise a modified sugar moiety and/or a modified nucleobase.

Constrained Ethyl Nucleoside (cEt)

[0088] As used herein, “constrained ethyl nucleoside” or “cEt” means a nucleoside comprising a bicyclic sugar moiety comprising a 4'-CH(CH₃)-O-2'bridge.

Locked Nucleic Acid Nucleoside (LNA)

[0089] As used herein, “locked nucleic acid nucleoside” or “LNA” means a nucleoside comprising a bicyclic sugar moiety comprising a 4'-CH₂-O-2'bridge.

2'-Substituted Nucleoside

[0090] As used herein, “2'-substituted nucleoside” means a nucleoside comprising a substituent at the 2'-position other than H or OH. Unless otherwise indicated, a 2'-substituted nucleoside is not a bicyclic nucleoside.

2'-deoxynucleoside

[0091] As used herein, “2'-deoxynucleoside” means a nucleoside comprising 2'-H furanosyl sugar moiety, as found in naturally occurring deoxyribonucleosides (DNA). In certain embodiments, a 2'-deoxynucleoside may comprise a modified nucleobase or may comprise an RNA nucleobase (e.g., uracil).

Oligonucleotide

[0092] As used herein, “oligonucleotide” means a compound comprising a plurality of linked nucleosides. In certain embodiments, an oligonucleotide comprises one or more unmodified ribonucleosides (RNA) and/or unmodified deoxyribonucleosides (DNA) and/or one or more modified nucleosides.

Terminal Group

[0093] As used herein, “terminal group” means one or more atoms attached to either, or both, the 3' end or the 5' end of an oligonucleotide. In certain embodiments a terminal group is a conjugate group. In certain embodiments, a terminal group comprises one or more terminal group nucleosides.

Conjugate

[0094] As used herein, “conjugate” means an atom or group of atoms bound to an oligonucleotide or oligomeric compound. In general, conjugate groups modify one or more properties of the

compound to which they are attached, including, but not limited to pharmacodynamic, pharmacokinetic, binding, absorption, cellular distribution, cellular uptake, charge and/or clearance properties.

Conjugate Linking Group

[0095] As used herein, “conjugate linking group” means any atom or group of atoms used to attach a conjugate to an oligonucleotide or oligomeric compound.

Splice Switching Oligonucleotide (SSO)

[0096] As used herein, “splice switching oligonucleotide” or “SSO” means a compound comprising or consisting of an oligonucleotide at least a portion of which is complementary to a target nucleic acid to which it is capable of hybridizing, resulting in at least one change in the splicing pattern of the targeted pre-mRNA.

[0097] A splice switching oligonucleotide could also be termed splice switching antisense oligomer (SSO).

mRNA

[0098] As used herein, “mRNA” means an RNA molecule that encodes a protein.

Pre-mRNA

[0099] As used herein, “pre-mRNA” means an RNA transcript that has not been fully processed into mRNA. Pre-mRNA includes one or more introns.

Target Pre-mRNA

[0100] As used herein, the term “target pre-mRNA” means a nucleic acid molecule to which an SSO hybridizes.

Change in the Splicing Pattern of the Targeted Pre-mRNA

[0101] As used herein, “a change in the splicing pattern of the targeted pre-mRNA” means a change in the pre-mRNA splicing process resulting in insertion of a proportion, for instance corresponding to a pseudoexon or a proportion thereof, into the produced mRNA when compared to the reference nucleotide sequence of the targeted pre-mRNA.

Transcript

[0102] As used herein, “transcript” means an RNA molecule transcribed from DNA. Transcripts include, but are not limited to mRNA, pre-mRNA, and partially processed RNA.

Targeting and Targeted to

[0103] As used herein, “targeting” or “targeted to” means the association of an SSO to a particular target nucleic acid molecule or a particular region of a target nucleic acid molecule. An SSO targets a target nucleic acid if it is sufficiently complementary to the target nucleic acid to allow hybridization under physiological conditions.

Nucleobase Complementarity and Complementarity

[0104] As used herein, “nucleobase complementarity” or “complementarity” when in reference to nucleobases means a nucleobase that is capable of base pairing with another nucleobase. For example, in DNA, adenine (A) is complementary to thymine (T). For example, in RNA, adenine (A) is complementary to uracil (U). In certain embodiments, complementary nucleobase means a nucleobase of an SSO that is capable of base pairing with a nucleobase of its target nucleic acid. For example, if a nucleobase at a certain position of an SSO is capable of hydrogen bonding with a nucleobase at a certain position of a target nucleic acid, then the position of hydrogen bonding between the oligonucleotide and the target nucleic acid is considered complementary at that nucleobase pair. Nucleobases comprising certain modifications may maintain the ability to pair with a counterpart nucleobase and thus, are still capable of nucleobase complementarity.

[0105] As used herein, “complementary” in reference to oligomeric compounds (e.g., linked nucleosides, oligonucleotides, or nucleic acids) means the capacity of such oligomeric compounds or regions thereof to hybridize to another oligomeric compound or region thereof through nucleobase complementarity under stringent conditions. Complementary oligomeric compounds need not have nucleobase complementarity at each nucleoside. Rather, some mismatches are

tolerated. In certain embodiments, complementary oligomeric compounds or regions are complementary at 70% of the nucleobases (70% complementary). In certain embodiments, complementary oligomeric compounds or regions are 80% complementary. In certain embodiments, complementary oligomeric compounds or regions are 90% complementary. In certain embodiments, complementary oligomeric compounds or regions are 95% complementary. In certain embodiments, complementary oligomeric compounds or regions are 100% complementary. In another embodiment, the oligomeric compounds comprise up to 3 mismatches, such as up to 2 or 1 mismatches. Preferably, no mismatches are present.

Hybridization

[0106] As used herein, “hybridization” means the pairing of complementary oligomeric compounds (e.g., an SSO and its target nucleic acid). While not limited to a particular mechanism, the most common mechanism of pairing involves hydrogen bonding, which may be Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding, between complementary nucleobases.

Motif

[0107] As used herein, “motif” means a pattern of chemical modifications in an oligomeric compound or a region thereof. Motifs may be defined by modifications at certain nucleosides and/or at certain linking groups of an oligomeric compound.

[0108] As used herein, “nucleoside motif” means a pattern of nucleoside modifications in an oligomeric compound or a region thereof. The linkages of such an oligomeric compound may be modified or unmodified. Unless otherwise indicated, motifs herein describing only nucleosides are intended to be nucleoside motifs. Thus, in such instances, the linkages are not limited.

[0109] As used herein, “sugar motif” means a pattern of sugar modifications in an oligomeric compound or a region thereof.

[0110] As used herein, “linkage motif” means a pattern of linkage modifications in an oligomeric compound or region thereof. The nucleosides of such an oligomeric compound may be modified or unmodified. Unless otherwise indicated, motifs herein describing only linkages are intended to be linkage motifs. Thus, in such instances, the nucleosides are not limited.

Type of Modification

[0111] As used herein, “type of modification” in reference to a nucleoside or a nucleoside of a “type” means the chemical modification of a nucleoside and includes modified and unmodified nucleosides. Accordingly, unless otherwise indicated, a “nucleoside having a modification of a first type” may be an unmodified nucleoside.

Differently Modified

[0112] As used herein, “differently modified” means chemical modifications or chemical substituents that are different from one another, including absence of modifications. Thus, for example, an MOE nucleoside and an unmodified DNA nucleoside are “differently modified,” even though the DNA nucleoside is unmodified. Likewise, DNA and RNA are “differently modified,” even though both are naturally occurring unmodified nucleosides. Nucleosides that are the same but comprise different nucleobases are not differently modified. For example, a nucleoside comprising a 2'-OMe modified sugar and an unmodified adenine nucleobase and a nucleoside comprising a 2'-OMe modified sugar and an unmodified thymine nucleobase are not differently modified.

MaxEnt Score

[0113] The MaxEnt score is a score known to the skilled person that accounts for adjacent as well as non-adjacent dependencies between positions within the splice site, using a maximum entropy principle to identify optimal splice sites. A high score indicates a high probability of a functionally strong splice site, but splice sites with lower scores may be functional through activation by splicing factors bound to the pre-mRNA at ESE or ISE motifs. Likewise, a splice site with a high score may be functionally repressed by nearby or overlapping ESS or ISS motifs binding inhibitory splicing factors.

[0114] MaxEnt score according to the present invention is determined using the program “MaxEntScan” version 20 Apr. 2004. The same software can be used to determine: [0115] 5' splice site scores: [0116] (http://hollywood.mit.edu/burgelab/maxent/Xmaxentscan_scoreseq.html); [0117] and [0118] 3' splice site scores: [0119] (http://hollywood.mit.edu/burgelab/maxent/Xmaxentscan_scoreseq_acc.html) [0120] Determination of MaxEnt score is further described in Gene Yeo and Christopher B Burge (J Comput Biol. 2004; 11(2-3):377-94).

[0121] The MaxEnt score is a specific value which can only be determined in one way.

[0122] Albeit the inclusion of MaxEnt scores significantly improves the overall predictability and is preferred, in specific embodiments of the invention, the MaxEnt score is optional.

Pseudoexon Identification

[0123] Pseudoexons are identified with precise genomic coordinates of the 3' splice site and the 5' splice site using a double-junction approach. In this approach, RNA sequencing fragments are filtered to retain only those with evidence of at least two splicing junctions. Exon coordinates can be extracted from the mapped reads, allowing for a gap of a certain length in the middle of the fragment where there is no direct sequence. The exons which are supported by a splicing junction at both ends in the same fragment are classified by comparing to a known gene annotation, and novel pseudoexons can be identified as exons that overlap introns, but not any existing exons.

[0124] The “Sweet Spot region” is defined as the region from +9 to +39 downstream of the 5' splice site of a pseudoexon, both positions included. Pseudoexons that can be activated by an SSO binding to a region within the Sweet Spot region is identified by the following parameters:

Pseudoexon Parameters/Criteria:

[0125] Pseudoexon length <160 nt; [0126] Pseudoexon length >30 nt; [0127] The last 3 nt of the pseudoexon are different from TAG; [0128] Donor splice site has a MaxEnt score 4.33; [0129] Donor splice site has a MaxEnt score 10.06; and [0130] Acceptor splice site has a MaxEnt score 3.63;

Sequence Region +9 to +39 Downstream to the 5' Splice Site of Said Pseudoexon

Parameters/Criteria:

[0131] Total pyrimidines ≤ 20 ; [0132] Total thymidine bases ≤ 12 ; [0133] Total thymidine bases ≥ 4 ; [0134] Total guanine bases ≤ 12 ; [0135] Maximum length of thymidine polymer ≤ 4 ; [0136] Maximum length of pyrimidine polymer ≤ 10 ; [0137] Minimum length of purine polymer ≥ 3 ; and [0138] Maximum number of guanine polymers of at least 3 nt length 2.

[0139] In an RNA sequence, the presence of a uracil may be considered equivalent to the presence of a thymidine at the same position in the corresponding DNA sequence. As such, the presence of a thymidine in the DNA sequence may also be considered equivalent to a uracil in the same position in the corresponding RNA sequence. The criteria covering sequences with thymidines and pyrimidines are therefore identical to equivalent criteria for sequences with uracil and pyrimidines when the analyzed sequence is an RNA sequence.

Identifying Sequences and Producing SSOs

[0140] As outlined above the present invention relates to the identification of sequence parameters in a gene comprising a pseudoexon, which can be used to determine if it is possible (with high probability) to get the pseudoexon incorporated in the mature mRNA using a splice switching oligonucleotide (SSO) (see also example 1 and corresponding figures). Thus, an aspect of the invention relates to a method for identifying SSOs able to modulate expression of a target protein in a cell by promoting incorporation of a pseudoexon into the mature mRNA upon binding to the pre-mRNA in the region +9 to +39 downstream to the 5' splice site of said pseudoexon, the method comprising: [0141] a) providing one or more gene sequences comprising one or more identified pseudoexons; [0142] b) determining for the one or more gene sequences; if the pseudoexon meets the following criteria: [0143] a pseudoexon length <160 nt; [0144] a pseudoexon length >30 nt; [0145] the last 3 nt of the pseudoexon are different from TAG; [0146] the donor splice site has a

MaxEnt score ≥ 4.33 ; [0147] the donor splice site has a MaxEnt score ≤ 10.06 ; and [0148] the acceptor splice site has a MaxEnt score ≥ 3.63 ; [0149] c) determining for the one or more gene sequences if the sequence region +9 to +39 downstream to the 5' splice site of said pseudoexon (3) meets the following criteria: [0150] total pyrimidines ≤ 20 ; [0151] total thymidine bases ≤ 12 ; [0152] total thymidine bases ≥ 4 ; [0153] total guanine bases ≤ 12 ; [0154] maximum length of thymidine polymer ≤ 4 ; [0155] maximum length of pyrimidine polymer ≤ 10 ; [0156] maximum length of purine polymer ≥ 3 ; and [0157] maximum number of guanine polymers of at least 3 nt length ≤ 2 ,

wherein, if one or more gene sequences meet the criteria according to point b) and point c), said region +9 to +39 downstream to the 5' splice site of said pseudoexon is considered a target for an SSO able to, in vivo, hybridize to the pre-mRNA (2) of said gene within the region +9 to +39 downstream to the 5' splice site of said pseudoexon (3); and resulting in said pseudoexon (3) becoming part of the mature mRNA to a greater extent compared to corresponding pre-mRNA not contacted with the SSO (1);

and

optionally, producing said SSO, optionally for use as a medicament.

[0158] As outlined in example 2, the included selection parameters (criteria) can discriminate between genes where SSOs can be used (Table 1 in example 2) and genes where the SSOs will not be able to incorporate the pseudoexon (Table 2 in example 2).

[0159] In an embodiment, the invention is computer-implemented (except for the optional step of producing said SSO, optionally for use as a medicament).

[0160] In another embodiment, the one or more gene sequences comprising one or more identified pseudoexons are provided in the form of a database or on another digital storage mean.

[0161] In an embodiment, the one or more gene sequences comprising one or more identified pseudoexons are gene sequences, which contains disease-causing genes, such as genes encoding dominant negative proteins, such as characterized by increased expression or altered function of the gene, or genes.

[0162] In another embodiment the pseudoexons is in a gene where decreased level of normal functional gene product has a therapeutic benefit, such as genes associated with cancer, diabetes, inflammation, neurodegenerative or neurological disorders, tissue degeneration, tissue fibrosis and chirosis, metabolic conditions, chronic liver disease and inherited retinal dystrophies (IRDs).

[0163] In yet another embodiment, the one or more gene sequences comprising one or more identified pseudoexons are gene sequences, which contains disease-causing genes, such as genes encoding proteins causing/enhancing/influencing diseases such as cancer, diabetes, inflammation, neurodegenerative or neurological disorders, tissue degeneration, tissue fibrosis and chirosis, metabolic conditions, chronic liver disease and inherited retinal dystrophies (IRDs). The disease may be due to enhanced expression.

[0164] In yet an embodiment, the one or more gene sequences comprising one or more identified pseudoexons are gene sequences, which are not only known to cause inherited disease(s).

[0165] In a preferred embodiment, SSOs are produced against an identified region +9 to +39 downstream to the 5' splice site of said pseudoexon, which region meets the above outlined criteria. Examples 3-10 show specific examples of the effect of SSOs against identified target sequences.

[0166] In an embodiment, said produced SSO comprises a sequence, which is complementary or substantially complementary to a region +9 to +39 downstream to the 5' splice site of said pseudoexon (3), such as within the region +11 to +35 downstream to the 5' splice site. In the example section (examples 2-8), SSOs which are complementary to position +11 to +35 have been used (25 nt long). In examples 9 and 10 specific optimization of the target region for the SSO has been further optimized for the two genes HIF1A and TRPM7. In examples 3 and 13 specific optimization of the target region for the SSO has been further optimized for the two genes SMAD2 and LRRK2.

[0167] In another embodiment, said produced SSO comprises a sequence which is substantially complementary to the region +9 to +39 downstream to the 5' splice site of said pseudoexon (3), and comprises at the most 3 mismatches, such as at the most 2 mismatches or such as at the most 1 mismatch.

[0168] In an embodiment, said produced SSO comprises a sequence which is complementary to a region +9 to +39 downstream to the 5' splice site of said pseudoexon (3) such as within the region +11 to +35 downstream to the 5' splice site. In the examples, SSOs targeting position +11 to +35 were tested.

[0169] In an embodiment, the complementary region being in the range 9-31 nucleotides, such as 15-30, such as 15-25 or such as 9-15, or such as 15-30, such as 20-25. If e.g. LNA are used or other high-binding nucleotides, the length of the SSO may be in the shorter ranges.

[0170] The SSOs may be able to modulate expression of the target protein in different ways. Thus, in an embodiment, hybridization of the SSO to the pre-mRNA in vivo results in: [0171] decreasing the level of mRNA encoding the functional protein (4); and/or [0172] decreasing expression of the functional protein (6); and/or [0173] loss of function of the functional protein (5); and/or [0174] a new function of the functional protein (6); and/or [0175] mis-localization of the protein. [0176] mis-localization of the mRNA encoding the functional protein.

[0177] In yet an embodiment, the one or more gene sequences from step a) causes a disorder or condition characterized by increased expression or altered function of the gene. In an embodiment, the disorder or condition is an autosomal dominant negative disorder.

[0178] In yet an embodiment, the one or more gene sequences from step a) is therapeutically beneficial when the level of normal functional gene product is decreased. In an embodiment, the disorder or condition is not directly associated with a disease-causing gene.

[0179] In an embodiment, said SSO has a length in the range 9-100 nucleotides, such as 9-50 nucleotides, preferably in the range 9-40 nucleotides and more preferably in the range 15-31 nucleotides or 15-25 nucleotides.

[0180] In an embodiment, said SSO comprises a sequence which is complementary or substantially complementary to a polynucleotide in the pre-mRNA, wherein said sequence has a length in the range 9-31 nucleotides, such as 15-25 nucleotides, preferably the sequence is complementary at a range of 9-31 nucleotides, such as 9-20, or such as 20-31 nucleotides, such as 25-31 nucleotides.

[0181] In a preferred embodiment, said produced SSO comprises one or more artificial nucleotides, such as sugar-modified nucleotides.

[0182] In another preferred embodiment, the SSO does not mediate RNase H mediated degradation of the mRNA in vivo.

[0183] In an embodiment, at least one modified sugar moiety is a 2'-substituted sugar moiety.

[0184] In an embodiment, said 2'-substituted sugar moiety has a 2'-substitution selected from the group consisting of 2'-O-Methyl (2'-OMe), 2'-fluoro (2'-F), and 2'-O-methoxyethyl (2'-MOE).

[0185] In an embodiment, said 2'-substitution of said at least one 2'-substituted sugar moiety is a 2'-O-methoxyethyl (2'-MOE).

[0186] In an embodiment, at least one modified sugar moiety is a bicyclic sugar moiety.

[0187] In an embodiment, at least one bicyclic sugar moiety is a locked nucleic acid (LNA) or constrained ethyl (cEt) nucleoside.

[0188] In an embodiment, at least one sugar moiety is a sugar surrogate.

[0189] In an embodiment, at least one sugar surrogate is a morpholino.

[0190] In an embodiment, at least one morpholino is a modified morpholino.

[0191] In an embodiment, the SSO comprises at least one internucleoside N3' to P5' phosphoramidate diester linkage.

[0192] In an embodiment, the modified oligonucleotide comprises at least one internucleoside phosphorothioate linkage.

[0193] In an embodiment, all internucleoside linkages are phosphorothioate.

[0194] In an embodiment, the SSO is conjugated to delivery elements, such as selected from the group consisting of Gal-Nac, (poly-)unsaturated fatty acids (such as oleoyl and linolenoyl), anisamide, anandamide, folic acid (FolA), carbachol, estrone, Retro-1, phospholipids, α -tocopherol (α -TP), cholesterol, squalene (SQ), unbranched fatty acids (such as lauroyl, myristoyl, palmitoyl, stearoyl, and docosanoyl), and cell penetrating peptides.

[0195] In an embodiment, the one or more gene sequences comprising one or more identified pseudoexons are involved in a disease or disorder selected from the group consisting of cancer, Inflammatory diseases, Neurodegenerative or neurological diseases, Metabolic conditions, Chronic liver disease and Inherited retinal dystrophies (IRDs).

[0196] In an embodiment, the Chronic liver disease is nonalcoholic fatty liver disease.

[0197] In an embodiment, said cancer is selected from the group consisting of, brain cancer, glioblastoma, lung cancer, colorectal cancer, skin cancer, pancreas cancer, bladder cancer, liver cancer, breast cancer, eye cancer and prostate cancer.

[0198] In yet an embodiment, said cancer is a haematological cancer, such as selected from the group consisting of multiple myeloma, acute myeloblastic leukemia, chronic myelogenic leukemia, acute lymphoblastic leukemia and chronic lymphocytic leukemia.

[0199] As outlined in the example section, the inventing team has identified a number of clinical relevant genes comprising pseudoexons, which can be incorporated in the mature mRNA, thereby inactivating/inhibiting/altering the function of the expressed (disease-causing) protein).

[0200] Thus, in an embodiment, said SSO is complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0201] a nucleic acid according to any of any of SEQ ID NO's: 106 (RNF115), 1-26, 79-105, 107-125 and 137-201; or [0202] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0203] a nucleic acid sequence having at least 90% sequence identity to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201.

[0204] In a preferred embodiment, said produced SSO is complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0205] a nucleic acid according to any of SEQ ID NO's: 79-100; or [0206] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 79-100; or [0207] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 79-100.

[0208] In a more preferred embodiment, said produced SSO is complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0209] a nucleic acid according to any of SEQ ID NO's: 79-85; or [0210] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 79-85; or [0211] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 79-85.

[0212] Based on the provided gene sequence data in the example section, the skilled person could easily translate this information into specific SSO sequences, e.g. by using the sections underlined in the tables and designing SSOs complementary thereto.

[0213] In an embodiment, the SSO is selected from the group consisting of: [0214] a nucleic acid according to any of SEQ ID NO's: 127-136; or [0215] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 127-136; or [0216] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 127-136.

[0217] Preferably, the nucleic acid is SEQ ID NO: 127 or 128 or 133 or 136. As shown in examples 9 and 10, these SSOs have been optimized within the Sweet Spot region. Thus, by moving the binding region e.g. just 1 or 2 positions the efficiency can surprisingly be increased even further. More preferably, the nucleic acid is SEQ ID NO: 128 or 136.

[0218] In another preferred embodiment, the SSO is complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0219] a nucleic acid according to any of SEQ ID NO: 21 (targeting SMAD2), SEQ ID NO: 106 (targeting RNF115) and any of SEQ ID No's: 141, 142, 158, and 159 (targeting LRRK2); [0220] a nucleic

acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 21, 106, 141, 142, 158, and 159; or [0221] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 21, 106, 141, 142, 158, and 159.

[0222] Example 3 shows data on SMAD2 targeting (see also example 11).

[0223] Example 12 shows data in RNF115 targeting.

[0224] Example 13 shows data on LRRK2 targeting, including allele specific targeting.

[0225] In yet another preferred embodiment, the SSO is complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0226] a nucleic acid according to any of SEQ ID NO: 100, 114, 118, 150 and 151; [0227] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 100, 114, 118, 150 and 151; or [0228] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 100, 114, 118, 150 and 151.

[0229] In yet another preferred embodiment, the SSO is complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0230] a nucleic acid according to any of SEQ ID NO: 12, 24 og 26; [0231] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 12, 24 og 26; or [0232] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 12, 24 og 26.

[0233] As outlined in the example section, different genes have already been targeted using the selection criteria according to the invention (Example 2, Table 1) or has been identified as targeting sequences using the selection criteria according to the invention (Example 8, Table 3, Example 11, tables 6-7). Thus, in an embodiment, the produced SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1, Table 3, Table 6 and Table 7 (Sweet Spot region). The genes listed in Table 1 or Table 3 or Table 6-7, may be specifically preferred in relation to certain diseases, as outlined below. Again, Sweet Spots for the SSOs are outlined in Table 1 and Table 3 and Tables 6-7. Table 2 shows pseudoexon sequences for which the criteria according to the invention is not fulfilled and which are not functional sites for SSOs.

Cancer Out of Frame (NMD):

[0234] The following genes could be relevant to target in relation to cancer treatment: TXNRD1, SLC7A11, STAT5B, MAPKAPK5, ZYG11A, ROCK1, MCCC2, SMYD2, DIAPH3, COPS3, SNX5, YBX1, CHD1L, PTPN11, UBAP2L, RNF115, HGS, TLK1, WWTR1, HMGCS1, SND1, THOC2, ORC1, TAF2, HIF1A, TRPM7, CPPS1, LRP6, MELK, TTBK2, TTK, ITGBL1, ROCK2, TASP1, FLT1, KNTC1, SMC1A, ZNF558, PMPCB and DBI.

[0235] Thus, in an embodiment, the produced SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Tables 6-7, wherein the gene is selected from the group consisting of TXNRD1, SLC7A11, STAT5B, MAPKAPK5, ZYG11A, ROCK1, MCCC2, SMYD2, DIAPH3, COPS3, SNX5, YBX1, CHD1L, PTPN11, UBAP2L, RNF115, HGS, TLK1, WWTR1, HMGCS1, SND1, THOC2, ORC1, TAF2, HIF1A, TRPM7, CPPS1, LRP6, MELK, TTBK2, TTK, ITGBL1, ROCK2, TASP1, FLT1, KNTC1, SMC1A, ZNF558, PMPCB and DBI; for use in the treatment of cancer. In a preferred embodiment the gene is RNF115 (see example 12).

Cancer in Frame:

[0236] The following genes could be relevant to target in relation to cancer treatment: ROCK1, E2F3, LRIG2, HSPG2, SLC2A1, KNTC1, DIAPH3, FDFT1, THOC2 and SMC1A, DDR2, STAG2, TRPM7, LINGO2, RAP1GDS, BUD1, CD44, CDKL5, RNF115, UBAP2L, ZNF558, RBPJ, EFEMP1, and FLT1.

[0237] Thus, in an embodiment, the produced SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Tables 6-7, wherein the gene is selected from the group consisting of ROCK1, E2F3, LRIG2, HSPG2, SLC2A1, KNTC1, DIAPH3, FDFT1, THOC2, DDR2, STAG2, TRPM7, LINGO2, SMC1A, RAP1GDS, BUD1, CD44, CDKL5, RNF115, UBAP2L, ZNF558, RBPJ, EFEMP1, and FLT1; for use in the treatment

of cancer.

Neurological Disease (Out of Frame):

[0238] The following genes could be relevant to target in relation to Neurological diseases:

ROCK1, HTT, OGA, TMEM97, PICALM, LRRK2, UBAP2L, SMC1A, TTBK2.

[0239] Thus, in an embodiment, the produced SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Tables 6-7, wherein the gene is selected from the group consisting of ROCK1, HTT, OGA, TMEM97, PICALM, LRRK2, UBAP2L, SMC1A and TTBK2; for use in the treatment of a neurological disease. In a preferred embodiment the gene is LRRK2 (see example 13).

[0240] In an even more preferred embodiment, the neurological disease is selected from the group consisting of Alzheimer's disease, Amyotrophic lateral sclerosis, Friedreich's ataxia, Huntington's disease, Lewy body disease, Parkinson's disease and Spinal muscular atrophy.

[0241] Neurodegeneration is the progressive loss of structure or function of neurons, including death of neurons. Many neurodegenerative diseases—including amyotrophic lateral sclerosis, Parkinson's disease, Alzheimer's disease, Huntington's disease, and prion diseases—occur as a result of neurodegenerative processes.

Neurological Disease (in Frame):

[0242] The following genes could be relevant to target in relation to Neurological diseases:

ROCK1, E2F3, SLC2A13, ASIC1, TRPM7, LINGO2, LRIG2, LRRK2, UBAP2L, SMC1A, ATXN7, and CLCN1.

[0243] Thus, in an embodiment, the produced SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Tables 6-7, wherein the gene is selected from the group consisting of ROCK1, E2F3, SLC2A13, TRPM7, LINGO2, ASIC1, LRIG2, LRRK2, UBAP2L, SMC1A, ATXN7, and CLCN1; for use in the treatment of a neurological disease.

[0244] Again, in an even more preferred embodiment, the neurological disease is selected from the group consisting of Alzheimer's disease, Amyotrophic lateral sclerosis, Friedreich's ataxia, Huntington's disease, Lewy body disease, Parkinson's disease and Spinal muscular atrophy.

Diabetes:

[0245] The following genes could be relevant to target in relation to diabetes: TXNRD1 (Diabetes), DYRK1A (Diabetes) TRPM7 (Diabetes) and PHLPP1 (Diabetes and obesity).

[0246] Thus, in an embodiment, the produced SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Tables 6-7, wherein the gene is selected from the group consisting of TXNRD1, DYRK1A, TRPM7 and PHLPP1; for use in the treatment of a diabetes. In a preferred embodiment, diabetes is selected from type 1 diabetes and type 2 diabetes.

[0247] In another embodiment, the produced SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1, Table 3 and Tables 6-7, wherein the gene is selected from the group consisting of LINGO2, SMAD2, ORC1, DDR2, STAG2, TRPM7, HIF1A, HTT, TAF2, CSPP1, RN115, LRRK2, UBAB2L, LRP6, MELK, and KNTC1. These 16 genes all comprise pseudoexons matching all criteria, all activated by SSO located within the Sweet Spot region (see Table 1 and Table 3 and Tables 6-7) and with high therapeutic potential.

[0248] In an embodiment, the method is computer-implemented. Thus, the invention can be implemented by means of hardware, software, firmware or any combination of these. The invention or some of the features thereof can also be implemented as software running on one or more data processors and/or digital signal processors.

[0249] The individual elements of an embodiment of the invention may be physically, functionally and logically implemented in any suitable way such as in a single unit, in a plurality of units or as part of separate functional units. The invention may be implemented in a single unit, or be both physically and functionally distributed between different units and processors.

[0250] As mentioned above, the SSO may comprise one or more mismatches. An advantage of introducing such mismatches is that allele-specific targeting is possible. This may be relevant when you only want to target one allele of a gene.

[0251] Specific SSOs are listed in Table 4, 5 and 8 targeting the genes (pre-mRNA) of the listed genes. Thus, in an embodiment, the SSO is selected from the SSOs listed in Tables 4, 5 and 8.

Composition for Use

[0252] The SSOs identified by the method of the invention, can be used as medicaments for the treatment of different diseases. Thus, another aspect of the invention relates to a composition comprising a splice switching oligonucleotide (SSO) being complementary or substantially complementary to a target pre-mRNA (e.g. encoding a functional disorder-causing or disorder-influencing protein), said target pre-mRNA (2) comprising: [0253] a function-disabling pseudoexon comprising: [0254] at the 5'-end a 3' splice site; and [0255] at the 3'-end a 5' splice site;

wherein said SSO is complementary or substantially complementary to the target pre-mRNA at a region +9 to +39 downstream to the 5' splice site of said pseudoexon;

wherein, when said SSO, in vivo, hybridizes to the pre-mRNA within the region +9 to +39 downstream to the 5' splice site of said pseudoexon; said pseudoexon becomes part of the mature mRNA to a greater extent compared to corresponding pre-mRNA not contacted with the SSO; for use as a medicament.

[0256] Thus, the SSOs modulate expression of a target protein by promoting incorporation of a pseudoexon into the mature mRNA.

[0257] The modulation induced by the SSOs may influence the target protein in different ways. Thus, in an embodiment, hybridization of the SSO to the pre-mRNA results in: [0258] decreasing the level of mRNA encoding the functional protein; and/or [0259] decreasing expression of the functional protein; and/or [0260] loss of function of the functional protein; and/or [0261] a new function of the functional protein; and/or [0262] mis-localization of the protein; and/or [0263] mis-localization of the mRNA encoding the functional protein.

[0264] In an embodiment, the composition is for use in the treatment of a human subject having a disease or condition characterized by increased expression or altered function of the disorder-causing or disorder-influencing functional protein, or where decreased expression of the functional gene product is therapeutically beneficial.

[0265] In an embodiment, said SSO comprises a sequence which is complementary or substantially complementary to a polynucleotide in the pre-mRNA characterized by the parameters according to this invention.

[0266] In an embodiment, said SSO comprising a sequence which is substantially complementary to the polynucleotide in the pre-mRNA, comprises at the most 3 mismatches, such as at the most 2 mismatches or such as at the most 1 mismatch.

[0267] In an embodiment, said SSO comprises a sequence, which is complementary to a polynucleotide in the pre-mRNA according to the defined criteria.

[0268] In an embodiment, the region +9 to +39 relative to the 5' splice site of said pseudoexon comprises a splicing regulatory site.

[0269] In yet an embodiment, the splicing regulatory site is an Intronic Splicing Silencer (ISS) site.

[0270] In an embodiment, said SSO has a length in the range 9-100 nucleotides, such as 9-50 nucleotides, preferably in the range 9-40 nucleotides and more preferably in the range 9-31 nucleotides or 9-25 nucleotides.

[0271] In an embodiment, said SSO comprises a sequence which is complementary or substantially complementary to a polynucleotide in the pre-mRNA as defined above, wherein said sequence has a length in the range 9-31 nucleotides, such as 9-20 nucleotides, preferably the sequence is complementary at a range of 9-31 nucleotides, such as 9-20, or such as 20-31 nucleotides, such as 25-31 nucleotides.

[0272] In an embodiment, said SSO comprises one or more artificial nucleotides, such as sugar-modified nucleotides.

[0273] In an embodiment, the oligonucleotide does not mediate RNase H mediated degradation of the mRNA.

[0274] In an embodiment, at least one modified sugar moiety is a 2'-substituted sugar moiety. In an embodiment, said 2'-substituted sugar moiety has a 2'-substitution selected from the group consisting of 2'-O-Methyl (2'-OMe), 2'-fluoro (2'-F), and 2'-O-methoxyethyl (2'-MOE).

[0275] In an embodiment, said 2'-substitution of said at least one 2'-substituted sugar moiety is a 2'-O-methoxyethyl (2'-MOE).

[0276] In an embodiment, the at least one modified sugar moiety is a bicyclic sugar moiety.

[0277] In an embodiment, the at least one bicyclic sugar moiety is a locked nucleic acid (LNA) or constrained ethyl (cEt) nucleoside.

[0278] In an embodiment, the at least one sugar moiety is a sugar surrogate.

[0279] In an embodiment, said at least one sugar surrogate is a morpholino.

[0280] In an embodiment, said at least one morpholino is a modified morpholino.

[0281] In an embodiment, the SSO comprises at least one internucleoside N3' to P5' phosphoramidate diester linkage.

[0282] In an embodiment, the modified oligonucleotide comprises at least one internucleoside phosphorothioate linkages.

[0283] In an embodiment, all internucleoside linkages are phosphorothioate.

[0284] In the example section, the tested SSOs were 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety.

[0285] In an embodiment, the SSO is conjugated to delivery elements, such as selected from the group consisting of Gal-Nac, (poly-)unsaturated fatty acids (such as oleoyl and linolenoyl), anisamide, anandamide, folic acid (FolA), carbachol, estrone, Retro-1, phospholipids, α -tocopherol (α -TP), cholesterol, squalene (SQ), unbranched fatty acids (such as lauroyl, myristoyl, palmitoyl, stearoyl, and docosanoyl), and cell penetrating peptides.

[0286] The composition can be used in the treatment of specific diseases. Thus in an embodiment, the composition for use in the treatment or alleviation of a disease selected from the group consisting of cancer, Inflammatory diseases, Neurodegenerative or neurological diseases, Metabolic conditions, Chronic liver disease and Inherited retinal dystrophies (IRDs).

[0287] In an embodiment, [0288] the disease is cancer and the gene sequence is selected from the group consisting of ROCK1, TXNRD1, SLC7A11, STAT5B, MAPKAPK5, ZYG11A, MCCC2, SMYD2, DIAPH3, COPS3, SNX5, YBX1, CHD1L, PTPN11, UBAP2L, RNF115, HGS, TLK1, WWTR1, HMGCS1, SND1, THOC2, E2F3, LRIG2, HSPG2, SLC2A1, KNTC1, FDFT1, SMC1A, HIF1A, CSPP1, TRPM7, DDR2, STAG2, ORC1, TAF2, LRP6, MELK, TTBK2, TTK, ITGBL1, ROCK2, TASP1, FLT1, ZNF558, PMPCB, DBI, RAP1GDS, BUD1, CD44, CDKL5, ZNF558, RBPJ, EFEMP1, and FLT1. [0289] or [0290] the disease is an inflammatory disease and the gene sequence is selected from the group consisting of DDR2 TRPM7, SMAD2, and LRP6; [0291] or [0292] the disease is a neurodegenerative or neurological disease and the gene sequence is selected from the group consisting of ROCK1, OGA, TMEM97, PICALM, E2F3, SLC2A13, ASIC1, TRPM7, LRIG2, LRRK2, UBAP2L, SMC1A, TTBK2, ATXN7, CLCN1; [0293] or [0294] the disease is Chronic liver disease and the gene sequence is selected from the group consisting of SMAD2 and TRPM7, DDR2, HIF1A, and ROCK1, RAP1GDS1; [0295] or [0296] the disease is Diabetes and the gene sequence is selected from the group consisting of TXNRD1, DYRK1A, TRPM7 and PHLPP1.

[0297] Sweet Spot target sequences for some SSOs are provided in the example section (Table 1 and Table 3 and Table 6 and Table 7). Thus, in an embodiment, said composition for use comprises an SSO complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0298] a nucleic acid according to any of any of SEQ ID NO's: 106, 1-

26, 79-105, 107-125 and 137-201; or [0299] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0300] a nucleic acid sequence having at least 90% sequence identity to any of any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201;

[0301] In a preferred embodiment, said composition for use comprises an SSO complementary or substantially complementary to region within a nucleic acid selected from the group consisting of:

[0302] a nucleic acid according to any of SEQ ID NO's: 79-100; or [0303] a nucleic acid

comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 79-100; or [0304] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 79-100.

[0305] In a more preferred embodiment, said composition for use comprises an SSO

complementary or substantially complementary to region within a nucleic acid selected from the group consisting of: [0306] a nucleic acid according to any of SEQ ID NO's: 79-85; or [0307] a

nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 79-85; or [0308] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 79-85.

[0309] In an embodiment, the SSO is selected from the group consisting of: [0310] a nucleic acid according to any of SEQ ID NO's: 127-136; or [0311] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 127-136; or [0312] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 127-136.

[0313] In yet another preferred embodiment, the SSO is complementary or substantially

complementary to region within a nucleic acid selected from the group consisting of [0314] a

nucleic acid according to any of SEQ ID NO: 100, 114, 118, 150 and 151; [0315] a nucleic acid

comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 100, 114, 118, 150

and 151; or [0316] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 100, 114, 118, 150 and 151.

[0317] In yet another preferred embodiment, the SSO is complementary or substantially

complementary to region within a nucleic acid selected from the group consisting of [0318] a

nucleic acid according to any of SEQ ID NO: 12, 24 og 26; [0319] a nucleic acid comprising 1 or 2

or 3 substitutions when compared to any of SEQ ID NO's: 12, 24 og 26; or [0320] a nucleic acid

sequence having at least 90% sequence identity to any of SEQ ID NO's: 12, 24 og 26.

[0321] Preferably, the SSO is SEQ ID NO: 128 or 136. As shown in examples 9 and 10, these SSOs have been optimized within the Sweet Spot region. Thus, by moving the binding region e.g. just 1 or 2 positions the efficiency can surprisingly be increased even further.

Other Composition for Use

[0322] In yet an aspect, the invention relates to a composition for use as a medicament, said

composition comprising [0323] an SSO complementary or substantially complementary to region

within a nucleic acid selected from the group consisting of [0324] a nucleic acid according to any

of SEQ ID NO's: 79, or 1-26 or 80-125; or [0325] a nucleic acid comprising 1 or 2 or 3

substitutions when compared to any of SEQ ID NO's: 79, or 1-26 or 80-125; or [0326] a nucleic

acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 79, or 1-26 or 80-125;

or [0327] an SSO selected from the group consisting of: [0328] a nucleic acid according to any of

SEQ ID NO's: 127-136; or [0329] a nucleic acid comprising 1 or 2 or 3 substitutions when

compared to any of SEQ ID NO's: 127-136; or [0330] a nucleic acid sequence having at least 90%

sequence identity to any of SEQ ID NO's: 127-136.

[0331] Preferably, the SSO is SEQ ID NO: 128 or 136. As shown in examples 9 and 10, these SSOs have been optimized within the Sweet Spot region. Thus, by moving the binding region e.g. just 1 or 2 positions the efficiency can surprisingly be increased even further.

[0332] As shown in examples 9 and 10, the SSOs targeting TRPM7 and HIF1A have been optimized within the Sweet Spot region.

[0333] Similar, optimization data for SMAD2 and LRRK2 are shown in Examples 3 and 13

respectively.

[0334] Thus, by moving the binding region just 1 or a few positions, the efficiency can surprisingly be increased even further.

[0335] As also described above and in the example section, different genes have already been targeted using the selection criteria according to the invention (Table 1 and Table 6) or has been identified as targeting sequences using the selection criteria according to the invention (Table 3 and Table 7). Thus, in an embodiment, the SSO is complementary or substantially complementary to region within a nucleic acid to a SEQ ID NO as outlined in Table 1, Table 3, Table 6 and Table 7 (Sweet Spot region).

[0336] In an embodiment, the SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3, wherein the gene is selected from the group consisting of TXNRD1, SLC7A11, STAT5B, MAPKAPK5, ZYG11A, ROCK1, MCCC2, SMYD2, DIAPH3, COPS3, SNX5, YBX1, CHD1L, PTPN11, UBAP2L, RNF115, HGS, TLK1, WWTR1, HMGCS1, SND1, HIF1A, CSPP1, TAF2, ORC1, THOC2, LRP6, MELK, TTBK2, TTK, ITGBL1, ROCK2, TASP1, FLT1, KNTC1, SMC1A, ZNF558, PMPCB and DBI; for use in the treatment of cancer.

[0337] In another embodiment, the SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Table 6 and Table 7, wherein the gene is selected from the group consisting of ROCK1, E2F3, LRIG2, HSPG2, SLC2A1, KNTC1, DIAPH3, FDFT1, THOC2, SMC1A, DDR2, LINGO2, TRPM7, STAG2, RAP1GDS, BUD1, CD44, CDKL5, RNF115, UBAP2L, ZNF558, RBPJ, EFEMP1, FLT1; for use in the treatment of cancer.

[0338] In yet an embodiment, the SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Table 6 and Table 7, wherein the gene is selected from the group consisting of ROCK1, OGA, TMEM97, PICALM, LRRK2, UBAP2L, SMC1A, and TTBK2; for use in the treatment of a neurological disease.

[0339] In an even more preferred embodiment, the neurological disease is selected from the group consisting of Alzheimer's disease, Amyotrophic lateral sclerosis, Friedreich's ataxia, Huntington's disease, Lewy body disease, Parkinson's disease and Spinal muscular atrophy.

[0340] Neurodegeneration is the progressive loss of structure or function of neurons, including death of neurons. Many neurodegenerative diseases—including amyotrophic lateral sclerosis, Parkinson's disease, Alzheimer's disease, Huntington's disease, and prion diseases—occur as a result of neurodegenerative processes.

[0341] In an embodiment, the SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Table 6 and Table 7, wherein the gene is selected from the group consisting of ROCK1, E2F3, SLC2A13, LINGO2, TRPM7, ASIC1, LRIG2, LRRK2, UBAP2L, SMC1A, ATXN7, and CLCN1; for use in the treatment of a neurological disease.

[0342] Again, in an even more preferred embodiment, the neurological disease is selected from the group consisting of Alzheimer's disease, Amyotrophic lateral sclerosis, Friedreich's ataxia, Huntington's disease, Lewy body disease, Parkinson's disease and Spinal muscular atrophy.

[0343] In an embodiment, the SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Table 6 and Table 7, wherein the gene is selected from the group consisting of TXNRD1, DYRK1A, TRPM7 and PHLPP1; for use in the treatment of a diabetes. In a preferred embodiment, diabetes is selected from type 1 diabetes and type 2 diabetes.

[0344] In another embodiment, the SSO is complementary or substantially complementary to a SEQ ID NO as outlined in Table 1 and Table 3 and Table 6 and Table 7, wherein the gene is selected from the group consisting of LINGO2, SMAD2, ORC1, DDR2, STAG2, TRPM7, HIF1A, HTT, TAF2, CSPP1, RN115, LRRK2, UBAB2L, LRP6, MELK, and KNTC1. These 16 genes all comprise pseudoexons matching all criteria, all activated by SSO located within the Sweet Spot region (see Table 1 and Table 3 and Table 6 and Table 7) and with high therapeutic potential.

Compositions

[0345] In yet an aspect the invention relates to a [0346] a composition comprising an SSO complementary or substantially complementary to region within a nucleic acid selected from the group consisting of [0347] a nucleic acid according to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0348] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or [0349] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201;

or [0350] a composition comprising an SSO selected from the group consisting of: [0351] a nucleic acid according to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or [0352] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or [0353] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216.

[0354] In an aspect the SSO is selected from the group of SSOs listed in table 4 (see example 9), Table 5 (see example 10) and Table 8 (see example 11).

[0355] Such compositions may be used as medicaments as outlined above, e.g. for the treatment of the list of diseases outlined above.

Allele Specific Targeting

[0356] By using the method according to the invention, the inventing team has identified SNPs inside a Sweet Spot region in the LRRK2 pre-mRNA (see also example 13).

[0357] By carefully designing SSOs it is considered plausible that such allele-specific SSOs can be used for preferentially targeting the disease-causing pre-mRNA (from the disease causing allele), whereas the pre-mRNA from the “normally functioning” allele is unaffected (or less affected).

[0358] By using such method, it is therefore possible to maintain an amount of normal RNA and thus maintaining normal gene-function.

[0359] As can be seen in example 13, SEQ ID NO's: 141 and 142 will target one SNP specific allele, whereas SEQ ID NO's: 158 and 159 will target another SNP specific allele.

[0360] Thus, in an embodiment the composition according to the invention is administered to a subject who is heterozygous in the pre-mRNA region targeted by the SSO, resulting in the SSO having an increased binding affinity to pre-mRNA of one of the alleles, such as to provide an increased splice switching activity in said allele.

[0361] In another embodiment, the pre-mRNA encodes for LRRK2.

[0362] In yet an embodiment, the pre-mRNA encodes for LRRK2 and the subject is heterozygous at the rs17444202 position.

[0363] In yet another embodiment, the pre-mRNA encodes for LRRK2 and the subject is heterozygous for a disease causing mutation in LRRK2.

[0364] In yet another embodiment, the SSO is complementary or substantially complementary to a region within a nucleic acid selected from the group consisting of [0365] a nucleic acid according to any of SEQ ID NO's: 141, 142, 158 and 159; or [0366] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 141, 142, 158 and 159; or [0367] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 141, 142, 158 and 159;

or [0368] the SSO is selected from the group consisting of: [0369] a nucleic acid according to any of SEQ ID NO's: 208-216; or [0370] a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 208-216; or [0371] a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 208-216;

[0372] In an embodiment, the SSO promotes inclusion of a pseudo-exon to a greater extend of disease-causing allele compared to the other allele. Again, this allows for the presence of the normal mRNA to a higher extent.

[0373] Yet an aspect of the invention relates to a method for identifying a subject who is likely eligible for allele-specific targeting of a dysfunctional LRRK2 allele, the method comprising

[0374] determining in a DNA sample from said subject the allelic status for LRRK2; wherein if the subject is heterozygous for the rs17444202, said subject is eligible for allele-specific treatment according to the invention; or wherein if the subject is not heterozygous rs17444202, said subject is not eligible for allele-specific treatment according to the invention.

[0375] Allelic status may be determined by Sanger or Next Generation (NGS) sequencing or by mutation specific assay, like ARMS or Taq-man.

Computer-Implemented

[0376] In an aspect, the invention relates to a computer program product being adapted to enable a computer system comprising at least one computer having data storage means in connection therewith to control a method according to the one or more aspects of the invention, such as a computer program product comprising instructions which, when the program is executed by a computer, cause the computer to carry out [the steps of] the method of the invention.

[0377] This aspect of the invention is particularly, but not exclusively, advantageous in that the present invention may be accomplished by a computer program product enabling a computer system to carry out the operations of the apparatus/system of the aspects of the invention when down- or uploaded into the computer system. Such a computer program product may be provided on any kind of computer readable medium, or through a network.

[0378] The individual aspects of the present invention may each be combined with any of the other aspects and embodiments. These and other aspects of the invention will be apparent from the following description with reference to the described embodiments.

[0379] Although the present invention has been described in connection with the specified embodiments, it should not be construed as being in any way limited to the presented examples. The scope of the present invention is to be interpreted in the light of the accompanying claim set. In the context of the claims, the terms “comprising” or “comprises” do not exclude other possible elements or steps. Also, the mentioning of references such as “a” or “an” etc. should not be construed as excluding a plurality. The use of reference signs in the claims with respect to elements indicated in the figures shall also not be construed as limiting the scope of the invention. Furthermore, individual features mentioned in different claims, may possibly be advantageously combined, and the mentioning of these features in different claims does not exclude that a combination of features is not possible and advantageous.

[0380] All patent and non-patent references cited in the present application, are hereby incorporated by reference in their entirety.

[0381] The invention will now be described in further details in the following non-limiting examples.

EXAMPLES

Example 1—Schematic Explanation of the Invention

[0382] FIG. 1 and the corresponding figure legend outlines the basic principle behind the invention. Pseudoexons are intronic sequences flanked by a 3' and a 5' splice site. Pseudoexons are usually not recognized due to the normally low amounts of inclusion into the mRNA transcript and because the pseudoexon containing mRNA is often degraded by the nonsense mediated decay of mRNA (NMD) system. The Sweet Spot region is defined as the region +9 to +39 nucleotides downstream of the 5' splice site of a pseudoexon that obeys the criteria described (see e.g. example 2) in this application. Pseudoexon inclusion into the mRNA transcript can be activated and increased by employing SSOs complementary to the Sweet Spot region of pseudoexons fulfilling the criteria. Pseudoexon inclusion into the mRNA will modulate gene expression either at the mRNA level or protein level, by mislocalization, destabilization and degradation or alteration of protein function. 5'ss; 5' splice site, 3'ss; 3' splice site, SSO; splice shifting oligonucleotide.

[0383] FIG. 2 demonstrates how RNA-sequencing data can be used in detection of in vivo spliced double junctions for empirical detection of pseudoexons, which are included into the endogenous

transcript at low levels.

[0384] After mapping to the human genome, reads are filtered to retain only fragments containing at least two splicing junctions. The splicing junctions of the entire fragment are then assembled into an exon structure, allowing for an unmapped gap between reads in the fragment of up to 100 bp. Exons are then classified using known exon annotations to identify pseudoexons contained within introns. Pseudoexons that may be candidates for activation by SSOs binding to the Sweet Spot region can be identified by the criteria according to the present invention, after which highly therapeutically relevant pseudoexons can be identified in genes where a downregulation of expression or alteration of the functional gene product is medically relevant. Subsequently, SSO can be produced using standard synthesis.

Example 2—Identification of Selection Parameters for SSOs

Aim of Study

[0385] Directly targeting specific genes as part of inhibiting a disease causing mRNA or protein in association with disease. By activating pseudoexons in target genes of interest, the resulting mRNA product will either be degraded through the NMD pathway or mis-localized or destabilized or being translated to a protein, which is non-functional or being unstable or mis-localized or having an altered function. We aimed to use Splice-switching antisense oligonucleotides (SSOs) to include pseudoexons in the mRNA transcript of the targeted gene. This strategy is superior to existing therapeutics that target multiple proteins, as the risk of off-target effects is minimized when using sequence specific SSOs, which are modified to achieve increased stability and binding specificity to their targeted sequence in a primary RNA transcript. Here we aimed to identify pseudoexons that can be activated, so that they are spliced into the mRNA by employing SSOs that bind in the +9 to +39 region (coined the Sweet Spot region) downstream of the pseudoexon donor site (5'-splice site), and to establish the criteria delineating these pseudoexons from non-activated pseudoexons.

Materials and Methods

[0386] We used public RNA-sequencing data (Geuvadis, E-MTAB-2836, E-MTAB-513, GSE52946, and GSE124439) and mapped them with STAR after trimming for adapter contamination and poorquality bases with bbdduk. HeLa cells were seeded in 12-well plates and forward transfected at 60% confluence with 20 or 40 nM 2'-O-methyl SSOs with full phosphorothioate backbone using Lipofectamine RNAiMAX (invitrogen). A non-binding ctrl SSO (5'GCUCAUAUAGCUACUGCCAUGCUUG3') (SEQ ID NO: 126) was used as control. RNA was harvested after 48 hours using Trizol (Invitrogen) and chloroform to isolate the RNA, followed by precipitation with isopropanol. Complementary DNA (cDNA) was synthesized from 500 ng RNA using the High capacity cDNA kit (Applied Biosystems). Primers were designed to span at least one exon-exon junction of the neighboring exons flanking the pseudoexons of interest. PCR was carried out using TEMPase Hot Start DNA polymerase (ampliqon) and 1 µl cDNA per reaction. 0.5 pmol/µl of each primer was used. The PCR products were separated on a 2% Seakem LE (Lonza) TBE agarose gel, for 1 hour at 80V.

[0387] The Sweet Spot region is located +9 to +39 of the 5' splice site of the pseudoexon.

[0388] All SSOs were 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety (Produced by LGC Biosearch Technologies). SSOs were used targeting position +11 to +35 inside the Sweet Spot region for that gene (relative to the 5' splice site of the pseudoexon). Thus, the SSOs binds inside the Sweet Spot.

Results

[0389] In order to identify pseudoexons across multiple tissues, we collected public RNA-sequencing data representing many different cell types. After mapping to the human genome, we filtered all reads for fragments containing at least two splicing junctions. We then mapped the splicing junctions of the entire fragment, allowing for an unmapped gap between reads in the fragment of up to 100 bp. From this we compiled a non-degenerate list of fully spliced exons from which we extracted the unknown exons contained within introns. Using this double-junction

approach, we were able to identify fully spliced pseudoexons even when expressed at very low levels. We have tested 78 SSOs targeting the Sweet Spot region from +11 to +35 downstream of randomly selected pseudoexons identified in RNA sequencing data. 26 of these SSOs were able to increase pseudoexon inclusion into the mRNA transcript of the targeted gene (Table 1 below), whereas the remaining 52 SSOs had no effect on pseudoexon inclusion (Table 2 below).

TABLE-US-00001 TABLE 1 26 pseudoexons matching all criteria, all activated by an SSO located within the Sweet Spot region. Sweet Spot region is annotated by its genomic sequence (DNA). SEQ ID NO: Gene hg38 pseudoexon coordinates Sweet Spot sequence 1 AGO3 chr1: 36005718-36005858(+)

AAGTTTATAGGTTAAATATTTATTAAAGCCA 2 ARFGEF1 chr8: 67226752-67226866(-)
ATTGCACAGCCTTCTTCAAGGAGGAGTCCCC 3 ATXN1 chr6: 16432945-16433096(-)
TGAAGTGTGTAGTGCCTCAGAAAATATCAAG 4 COPS5 chr8: 67056866-67056932(-)
AATCCAGCCTACAGCCACCAAACTGGGTGGG 5 CSPP1 chr8: 67162175-67162312(+)
ATGGCACATGTGGAGAATCTTCCATGGTGTG 6 CTNND2 chr5: 11083880-11083955(-)
CAGCTCTGGGGTGCCTGCCCCCATGGGGGAA 7 DDR2 chr1: 162756536-162756638(+)
CAGGAAGGGAGTGTGGAATTACAGCCTTCTA 8 DGCR8 chr22: 20109208-20109297(+)
GTGCACCTGCCAGGGTTGAATCTGTAGGGCT 9 DMTF1 chr7: 87184819-87184892(+)
CAGGCAGCGGCAGTGTAGCTTTTCCACTGTC 10 DMTF1 chr7: 87167956-87168043(+)
ACATGAAAGCACATGAGCTGCTTAATGTTGG 11 EYA4 chr6: 133301647-133301696(+)
CGGAAGTGGGATTTGGACCTACCTTCAGAAC 12 HIF1A chr14: 61724196-61724230(+)
AATCAAGCTTAGTTTATGAAGGACTGAACA 13 HTT chr4: 3144777-3144843(+)
GAGCTCAGCTCCTAAGGATGTGCAGGGGCAG 14 HTT chr4: 3102605-3102748(+)
GGGGCACCAAAGTCTTCCCTGTCCCATCCCC 15 LINGO2 chr9: 28370207-28370262(-)
GGATAGATAGAGAGGATCCAGAGGCTAAACG 16 NHSL1 chr6: 138487729-138487783(-)
CAGGATATTGGAAGTCAGGTCCCATCTTCGT 17 NAA25 chr12: 112049482-112049611(-)
TTGACTATTGGCCTATTTTCTACTGGGCAA 18 ORC1 chr1: 52394553-52394615(-)
CAAAAGCCATCTTTATGAGCACTCAGGTGCT 19 RBM22 chr5: 150692025-150692147(-)
GTGAACTGACATTTGCTTAAGCCCCGTGTCA 20 SLC20A2 chr8: 42471066-42471201(-)
CTGGTGTACTAAGAATACAGCCACATTCCCC 21 SMAD2 chr18: 47868051-47868097(-)
GTCTTAGTTGTTGAAGCTAAGCGAGATGCAG 22 STAG2 chrX: 124037905-124037969(+)
ACAGTGTAGGGAGTTAAAAAGTGCCATGAAT 23 TAF2 chr8: 119779073-119779146(-)
ACGGAAAAGCGCCTTTCCCTCAGTTGTTCT 24 TRPM7 chr15: 50588190-50588250(-)
TCTTTAGTATATGCAGAAGTCAGAAAGTCAG 25 ZNF264 chr19: 57206109-57206184(+)
AGCACCTACTCTCAGTCCACTACATTGAGTC 26 ZNF558 chr19: 8814260-8814371(-)
TGCTCCCTGCAAGTACTAGGAATGCAGGCTG

Underlining indicates tested or preferred binding sequence of the SSO. Binding is to the corresponding pre-mRNA. hg38 Genome coordinates follow BED format; zero-based start co-ordinate and one-based end coordinate.

TABLE-US-00002 TABLE 2 52 pseudoexons not matching all criteria, and not activated by SSO located within the Sweet Spot region. SEQ ID NO: Gene hg38 pseudoexon coordinates Sweet Spot sequence (5' to 3') 27 ARFGEF1 chr8: 67260568-67260632(-) TAAGTTTACCATCTTATATGGACGTGGTGTG 28 ATXN1 chr6: 16442867-16443008(-) TGGATTTTTTGGTAGAGACGGGGTTTTGCCAT 29 ATXN1 chr6: 16488491-16488637(-) ACAGACCACTGAACCAAGAACCAAGAGAGAA 30 ATXN1 chr6: 16319358-16319397(-) CCTTTTCACATTGGCGTCTCTCACTTAGCGA 31 ATXN1 chr6: 16494736-16494865(-) CATCTATATGTTCAACTTACATTTTATTTTC 32 ATXN2 chr12: 111520393-111520646(-) CACTGTGCCTGGCCGGTTTTGTCTTCTAAGT 33 ATXN2 chr12: 111554866-111555074(-) TGGCTTCCTACCCCATTTATTTATACTTCAC 34 ATXN2 chr12: 111461139-111461305(-) TTCAGATCTTTTGAGCTAGAACAAAAAACA 35 ATXN2 chr12: 111529929-111530026(-) TGTTAAATCTGATGTTAATGATTATTTTCATC 36 ATXN2 chr12: 111482787-111482831(-) ATTATAGAGTTTAATCTTATTTTGAGGGCCT 37 ATXN2 chr12:

111508911-111509038(-) TCCTCAGCCTGTGTTTTAGCTTTCTAAATGT 38 CACNA1G
 chr17: 50578805-50578849(+) CAGCATGTGGGGAGAGGCGCGTCACTGGGAG 39
 CACNA1H chr16: 1217594-1217708(+) GCAGGGGCGCTCGGCCAGGGGCTCCGACCTC 40
 CEP128 chr14: 80568955-80569044(-) TTTCATAGGTAGGAATATTTAGGATATTTGG 41
 CEP128 chr14: 80527169-80527261(-) CACTGTGTGTAGCCTTCTTCTTTCTCTATGG 42
 CSPP1 chr8: 67102087-67102144(+) AGTTAGCTAATCATTTTCTCTGTTTATATTT 43
 CTNND2 chr5: 11119101-11119134(-) GTAAGACTGACATGCCTTCTACCCTCCAGAA 44
 ERAP2 chr5: 96902643-96902842(+) ACCCTCTAGTATAATATATGCCACATTAAAA 45
 FKBP5 chr6: 35671139-35671256(-) CACCATGCCCAACTAATTTTCTATTTTTTG 46 HTT
 chr4: 3098129-3098222(+) TACTGCTAAGTGGCATGTTTTGTTTTATGCT 47 HTT chr4:
 3234807-3234939(+) GA CTGGCCTGGGGTGTGGGAATCTAGGGCCT 48 HTT chr4:
 3124250-3124362(+) CCCCCATGAGAGCTGTGTCTTCAAACCTCTT 49 HTT chr4:
 3213591-3213736(+) GCTTTGTGGCAGAGAGGGGACTGGCACTTTG 50 HTT chr4:
 3218488-3218645(+) TATTGATCAGAACCCTTGTTTCAGATAACAT 51 HTT chr4: 3223670-
 3223869(+) CATTTGGTATTACACCAGGTTCCCTTAGGCA 52 HTT chr4: 3124250-
 3124305(+) TCTTCTGGCTGGGACATGGGATATATCCTGT 53 LINGO2 chr9: 28436349-
 28436392(-) ACATATGTCTATGTTTCAGTGCCAATAGTTAA 54 MAP2K2 chr19: 4110359-
 4110415(-) ATGTAGATGCCTTTGGTTTTGTTTTGTTTTTC 55 MAP2K2 chr19: 4109169-
 4109284(-) CCGCCTGCCGCACAGCGTCGTTTGCAAAACC 56 MECP2 chrX: 154094378-
 154094489(-) AGATGTGAACAGGTCCCTCTTCTTTGGGCTT 57 MTOR chr1: 11140517-
 11140666(-) CCTGTTAGGAACCGGGAGGCACAGCAGGAGG 58 MUC16 chr19: 8853333-
 8853363(-) ATTTTTAGTTATTTGAGAAATCTCCACGTTT 59 NGLY1 chr3: 25761678-
 25761708(-) ATATTTATGTTTTTAAGAAATTACCAAGCTG 60 NUBPL chr14: 31654070-
 31654134(+) CATGAATTTTTTGGTTTCCCATTGCATATAA 61 NAA25 chr12: 112045827-
 112045878(-) TATTTTTAGTTACTTTTTATTTTTGAGACCT 62 PCSK9 chr1: 55045413-
 55045449(+) ACTCGCTGAAGTGGGGGCAGGTAAAGAAGCC 63 PCSK9 chr1: 55055176-
 55055228(+) GCAGCCAAGACTCTGTTCAAGTTTGTGTGGG 64 PSMG2 chr18: 12705934-
 12706047(+) TAAATAGTCTCAAAGGTGGAGGAGGCCCCAG 65 PTBP3 chr9: 112300837-
 112300928(-) CACTGTGCCCGGGTTTTTGTACCCCATATAA 66 SCN1A chr2: 166007230-
 166007293(-) TTACCCCTTTTGCTACCTTTAATCCTTGCAC 67 SCN1A chr2: 165992896-
 165993078(-) CAAGATCATGGGGAGATGAAAGTAGCATCAA 68 SCN9A chr2: 166307149-
 166307277(-) CTGATATTGATGTGAAAAATTGATATTTTGG 69 SCN9A chr2: 166321439-
 166321515(-) CCCACCTTAGCTACTCTCAAGCAGCTGAGAC 70 SDCCAG8 chr1:
 243497867-243497959(+) CACTGTTGGGTTGTTTTTCTTTTGAAGTGTT 71 SNCA chr4:
 89771397-89771585(-) AGGCATGCTGGACAAATGGATTCACATGTGC 72 SNCA chr4:
 89726926-89726976(-) TCATTAAATGGTGCATCCGGATCAGAACCTA 73 SNCA chr4:
 89836461-89836529(-) CCAACCTTTTCTCTCACATAAAATCTGTCTGC 74 SNCA chr4:
 89836337-89836394(-) GGGTTAACAAGTGCTGGCGCGGGGTCCGCTA 75 SORT1 chr1:
 109397207-109397340(-) TCTCTGGGTCAGTACTTTCCGGGTGGGAGAG 76 SP100 chr2:
 230436381-230436405(+) ATCATGGGGCAATTCACCCATTCTGTTCTC 77 TMEM243
 chr7: 87206679-87206742(-) TAAACTGTGCACTGTTCTATGTAATATGATT 78 YIPF6
 chrX: 68506157-68506258(+) GATCCCATATACACCTTCAACAATTATCTA

Underlining indicates tested or preferred binding sequence of the SSO. Binding is to the corresponding pre-mRNA. hg38 Genome coordinates follow GFF/GTF format; one-based start and end coordinates.

[0390] Based on these results we established a set of criteria, which must be met in order for a pseudoexon to be included by an SSO targeting the Sweet Spot region. Using these criteria, we have selected new targets for pseudoexon inclusion in disease causing genes where pseudoexon inclusion has a high potential for therapeutic use in human diseases.

Criteria:

For the Pseudoexon:

[0391] Pseudoexon length <160 nt; [0392] Pseudoexon length >30 nt; [0393] The last 3 nt of the pseudoexon are different from TAG; [0394] Donor splice site has a MaxEnt score ≥ 4.33 ; [0395] Donor splice site has a MaxEnt score ≤ 10.06 ; and [0396] Acceptor splice site has a MaxEnt score ≥ 3.63 ;

For the Gene Sequences +9 to +39 Downstream to the 5' Splice Site of Said Pseudoexon (3):

[0397] Total pyrimidines ≤ 20 ; [0398] Total thymidine bases ≤ 12 ; [0399] Total thymidine bases ≥ 4 ;

[0400] Total guanine bases ≤ 12 ; [0401] Maximum length of thymidine polymer ≤ 4 ; [0402]

Maximum length of pyrimidine polymer ≤ 10 ; [0403] Minimum length of purine polymer ≥ 3 ; and

[0404] Maximum number of guanine polymers of at least 3 nt length ≤ 2 ;

Conclusion

[0405] Pseudoexons in target genes of interest can be activated as a mechanism for downregulation of a disease-causing protein. By filtering pseudoexons based on the criteria we have established, the selection of new target candidates will enable the discovery of novel therapeutic agents.

Example 3—SMAD2

Aim of Study

[0406] Chronic liver disease is characterized by inflammation and fibrosis of the liver. Through in silico analysis and in vivo experiments, we aimed at investigating the presence of pseudoexons in the TGF- β /Smad signaling pathway, which is important for tissue fibrosis (Inagaki et al, 2007).

Materials and Methods

[0407] We used our novel double-junction approach to identify fragments with fully spliced pseudoexons in publicly available RNA-seq data from the GEUVADIS consortium. HeLa cells, LX-2 and HepG2 cells were seeded in 12-well plates and forward transfected at 60% confluence with 40 nM SSO using Lipofectamine RNAiMAX (Invitrogen). SSO (targeting SEQ ID NO: 21; specific target sequence is underlined in SEQ ID NO: 21 in table 1) was 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety (Produced by LGC Biosearch Technologies). (See also table from example 2). The SSO is complementary to position +11 to +35 inside the Sweet Spot region for the SMAD2 gene (relative to the 5' splice site of the pseudoexon) (SEQ ID NO: 21). Thus, the SSOs binds inside the Sweet Spot. In order to determine the optimal SSOs targeting the Sweet Spot region, we tested several SSOs employing 25 nt long SSOs targeting the Sweet Spot region from +9 to +14 position downstream of the 5'ss of the PE (SEQ ID NO: 202-207 listed in table 8 in example 11).

[0408] A non-binding SSO (5'-GCUCAUAUGCUACUGCCAUGCUUG-3') (SEQ ID NO: 126) with similar modifications was used as a negative control. For experiments with TGF β stimulation, the cells were stimulated with 10 ng/ml TGF β (R&D systems) for 16 hours before RNA and protein harvest. RNA was harvested after 48 hours using Trizol (Invitrogen) and chloroform to isolate the RNA, followed by precipitation with isopropanol. Complementary DNA (cDNA) was synthesized from 500 ng RNA using the High capacity cDNA kit (Applied Biosystems). Primers were designed to span at least one exon-exon junction of the neighboring exons flanking the pseudoexons of interest. PCR was carried out using TEMPase Hot Start DNA polymerase (ampliqon) and 1 μ l cDNA per reaction. 0.5 pmol/ μ l of each primer was used. The PCR products were separated on a 2% Seakem LE (Lonza) TBE agarose gel, for 1 hour at 80V. Proteins were extracted by lysing the cells with okadaic acid to preserve phosphorylated proteins, benzamide treated, and the denatured proteins were separated on a 4-12% NuPage SDS-Page gel and analyzed by western blotting using antibodies against SMAD2, phosphoSMAD2 and actin for control. For the study of myofibroblast formation, LX-2 cells were grown in 96 well plates transfected with 20 nM SSO and incubated in the Incucyte instrument with images taken every 4 hours. The images were analyzed in ImageJ by making a mask for spherical (differentiated) cells.

Results

[0409] By examining fragments with fully spliced pseudoexons, we identified a pseudoexon located within intron 5 of the SMAD2 gene, encoding the signal transducer protein Mothers against

decapentaplegic homolog 2, which is involved in the TGF- β /Smad pathway. Inclusion of the pseudoexon into a mature mRNA results in the insertion of 46 nt into the coding region, causing a frame-shift. A stop-codon (UAA) is also located within the pseudoexon, potentially activating the NMD pathway, causing degradation of the transcript.

[0410] Transfection of SSOs (SEQ ID NO: 202-207 listed in table 8, example 11) showed that all mediate pseudoexon inclusion, and that SSO targeting from +11 and +12 are optimal in mediating pseudoexon inclusion into the SMAD2 transcript (results not shown).

[0411] Transfection of HeLa cells with an SSO complementary to position +11 to +35 downstream of the 5' splice site resulted in up to 90% inclusion of the pseudoexon (FIG. 8A). Transfection of the same SSO in LX-2 Hepatic stellate cells resulted in even higher pseudoexon inclusion with lower concentration of SSO (FIG. 8B). Protein levels from HepG2 cells show a high decrease in SMAD2 protein, as well as phosphorylated SMAD2 under TGF β stimulation (FIG. 8C). TGF β stimulation of LX-2 cells resulted in differentiation of the cells into myofibroblast, which was reduced by transfection with the +11 SSO (FIG. 8D).

Conclusion

[0412] The normal function of the SMAD2 gene may be decreased by up to 90% using a specific SSO to increase inclusion of a pseudoexon thereby disrupting the function of the normal gene product, either through degradation of the transcript or expression of a truncated and non-functional protein. This has relevance in hepatic fibrosis and other disorders associated with increased TGF- β activity, of which SMAD2 is a positive regulator (Sysa et al, 2009). Additionally, SMAD2 downregulation may reduce growth of gliomas (Papachristodoulou et al, 2019) [0413] Inagaki Y, et al. Gut. 2007 February; 56(2):284-92. doi: 10.1136/gut.2005.088690 Sysa P, et al. 2009 September; 28(9):425-34. doi: [0414] 10.1089/dna.2009.0884 Papachristodoulou A, et al. Clin Cancer Res. 2019 Dec. 1; 25(23):7189-7201. doi: 10.1158/1078-0432.CCR-17-3024

Example 4—ORC1

Aim of Study

[0415] The origin of recognition complex (ORC) genes are involved in DNA replication and are expressed highly in hepatocellular carcinoma tumors (Wang et al, 2020). Low expression of ORC1 consistently indicates better prognosis compared to high ORC1 expression (Wang et al, 2020), while knockdown of ORC1 sensitises cancer cells, making them vulnerable to other anticancer treatments (Zimmerman et al, 2013). Through in silico analysis and in vivo experiments, we aimed at investigating the presence of pseudoexons in the ORC1 gene, which may be used to down-regulate the expression this gene.

Materials and Methods

[0416] We used our novel double-junction approach to identify fragments with fully spliced pseudoexons in publicly available RNA-seq data from the GEUVADIS consortium. HeLa cells were seeded in 12-well plates and forward transfected at 60% confluence with 40 nM SSO using Lipofectamine RNAiMAX (invitrogen).

[0417] SSO (targeting SEQ ID NO: 18; specific target sequence is underlined in SEQ ID NO: 18 in table 1) was 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety (Produced by LGC Biosearch Technologies). (See also table from example 2). The SSO is complementary to position +11 to +35 inside the Sweet Spot region for the ORC1 gene (relative to the 5' splice site of the pseudoexon). Thus, the SSOs binds inside the Sweet Spot.

[0418] A non-binding SSO (5'GCUCAAUAUGCUACUGCCAUGCUUG3') (SEQ ID NO: 126) with similar modifications was used as a negative control. RNA was harvested after 48 hours using Trizol (Invitrogen) and chloroform to isolate the RNA, followed by precipitation with isopropanol. Complementary DNA (cDNA) was synthesized from 500 ng RNA using the High capacity cDNA kit (Applied Biosystems). Primers were designed to span at least one exon-exon junction of the neighboring exons flanking the pseudoexons of interest. PCR was carried out using TEMPase Hot Start DNA polymerase (ampliqon) and 1 μ l cDNA per reaction. 0.5 pmol/ μ l of each primer was

used. The PCR products were separated on a 2% Seakem LE (Lonza) TBE agarose gel, for 1 hour at 80V.

Results

[0419] Using our double-junction approach to examine fragments with fully spliced pseudoexons, we identified a pseudoexon within intron 5 of the ORC1 gene, resulting in insertion of 62 nt into the pre-mRNA causing a frame-shift. Transfection of HeLa cells with an SSO complementary to position +11 to +35 downstream of the 5' splice site resulted in up to 45% more inclusion of the pseudoexon (data not shown).

Conclusion

[0420] The normal expression of the ORC1 gene product may be decreased by up to 45% using a specific SSO to increase inclusion of a pseudoexon disrupting function of the normal gene product. SSO-mediated down-regulation of ORC1 expression might therefore work in enhancing the anti-cancer effect of other drugs. [0421] Wang X K, et al. J Cancer. 2020 Jan. 20; 11(7):1869-1882. doi: 10.7150/jca.39163 Zimmerman, K. M., et al. Mol Cancer Res, 2013. 11(4): p. 370-80.

Example 5—LINGO2

Aim of Study

[0422] LINGO2 (Leucine-rich repeat and immunoglobulin-like domain-containing nogo receptor-interacting protein 2) is highly expressed in many tissues including intestinal tissues, brain and neurons (such as cortical neurons and dorsal root ganglion (DRG) neurons) (Guillemain et al. 2020). LINGO2 expression is increased in gastric cancer and this is associated with a poor prognosis and down regulation of LINGO2 decreases proliferation of gastric cancer cells (Jo et al. 2019). LINGO2 also negatively regulates motor neuron survival and motor neuron axonal length. This suggests that knock down or alteration of LINGO2 expression could be a new therapy against neuronal disorders and cancer.

Materials and Methods

[0423] We used the analysis pipeline (FIG. 2) to analyze publicly available RNA-seq data. HeLa cells were grown in RPMI and transfected with 20 nM SSO using Lipofectamine. After 24 hours, cells were harvested and RNA purified. RT-PCR was performed and the resulting product visualized on 2% agarose gel. For Incucyte experiments U251 cells were grown in 96-well plates transfected with SSOs at 5, 10, 20 or 40 nM and incubated in the incucyte instrument with images taken every 4.h.

Results

[0424] Based on the high LINGO2 expression in neurons and brain, we speculated if SSO based activation of the PE could be used for treating glioblastoma. By employing an SSO targeting the Sweet Spot region according to SEQ ID NO: 15 (specific target sequence is underlined in SEQ ID NO: 15 in table 1) we demonstrated a high level of inclusion of the chr9:28370208-28370262 (–) LINGO2 pseudoexon in U251 glioblastoma cells (FIG. 3A). This pseudoexon introduces 55 bp between exon 3 and 4 in the mRNA of the pseudoexon. When investigating U251 glioblastoma cells with this SSO, we observed a dose dependent reduction of growth and proliferation of U251 glioblastoma cells resulting from SSO treatment (FIG. 3B).

Conclusion

[0425] Accordingly, methods for treating cancer, such as gastric cancer or glioblastoma or for promoting survival of motor neurons and axonal growth of motor neurons by contacting human cells, such as cancer or neuronal cells with an SSO that causes inclusion of the pseudoexon into LINGO2 mRNA are provided herein.

Example 6—TAF2

Aim of Study

[0426] The TAF2 gene expresses the Tata-box binding protein associated factor 2, a subunit of the transcription factor II D complex involved in binding to promotor sequences to initiate transcription (Martinez et al. 1998). TAF2 exhibits copy number increases or mRNA

overexpression in 73% of high-grade serous ovarian cancers (HGSC) (Ribeiro et al. 2014) and is important in cancer.

Materials and Methods

[0427] We used the analysis pipeline (FIG. 2) to analyze publicly available RNA-seq data. NCI-H358 lung cancer cells were grown in RPMI and transfected with nM SSO using Lipofectamine. After 24 hours, cells were harvested and RNA purified. RT-PCR was performed and the resulting product visualized on 2% agarose gel. WST-1 assay was performed 48 hours post transfection. For Incucyte experiments NCI-H358 lung cancer cells were grown in 96-well plates transfected with SSOs at 10 nM and incubated in the incucyte instrument with images taken every 4.5 hours.

Results

[0428] Inclusion of the chr8:119779073-119779146(-) TAF2 pseudoexon will introduce 73 bp between exon 17 and 18 in the mRNA, resulting in a shifted reading frame with a premature stop codon in exon 18. TAF2 mRNA transcripts with inclusion of this pseudoexon are targets for nonsense-mediated mRNA decay (NMD), and increased pseudoexon inclusion induced by SSO treatment will therefore result in a reduction of expression of TAF2 mRNA. Moreover, if translated the pseudoexon included transcript will result in production of a severely truncated protein without normal TAF2 function.

[0429] By employing an SSO targeting the TAF2 Sweet Spot region according to SEQ ID NO: 23 (specific target sequence is underlined in SEQ ID NO: 23 in table 1) we demonstrated a high level of inclusion of chr8:119779073-119779146(-) TAF2 pseudoexon in NCI-H358 lung cancer cells (FIG. 4A). When investigating NCI-H358 lung cancer cells with this SSO, we observed reduction of growth and proliferation of by incucyte assay (FIG. 4B) and by WST-1 assay (FIG. 4C).

Conclusion

[0430] Employing the SSO targeting the TAF2 pseudoexon results in increased pseudoexon inclusion and reduced proliferation of cancer cells. SSO targeting TAF2 pseudoexon might therefore be candidates to be used in future anti-cancer therapy.

Example 7—HTT

Aim of Study

[0431] HTT encodes the huntingtin protein and is associated with the autosomal dominant neurodegenerative disorder, Huntington's disease, which is caused by unstable expansion of CAG trinucleotide repeats in the HTT gene that results in translation of a cytotoxic mutant protein with an abnormal polyglutamine tract. Downregulation of HTT has been studied as a potential strategy in treatment of Huntington's disease by reducing levels of mutant huntingtin. Inclusion of the chr4:3102605-3102748(+) HTT pseudoexon will introduce 143 bp between exon 3 and 4 in the mRNA, including an in-frame premature termination codon. HTT mRNA transcripts with inclusion of this pseudoexon is a target for nonsense-mediated decay (NMD), and increased pseudoexon inclusion induced by SSO treatment will result in a reduction of expression of HTT mRNA. Huntington disease is an autosomal dominant disease caused by a dominant negative tri-nucleotide repeat expansion in HTT mRNA. Inclusion of the PE will reduce levels of the dominant negative mRNA and may therefore be used as treatment.

Materials and Methods

[0432] We used the analysis pipeline (FIG. 2) to analyze publicly available RNA-seq data. HeLa cells were grown in RPMI and transfected with 20 nM SSO using Lipofectamine. After 24 hours, cells were harvested and RNA purified. RT-PCR was performed and the resulting product visualized on 2% agarose gel.

Results

[0433] By employing an SSO targeting the Sweet Spot region according to SEQ ID NO: 14 (specific target sequence is underlined in SEQ ID NO: 14 in table 1) we demonstrated a high level of inclusion of the chr4:3102605-3102748(+) HTT pseudoexon in HeLa cells which causes

degradation of HTT mRNA by the NMD system (Data not shown).

Conclusion

[0434] Employing the SSO targeting the HTT pseudoexon results in increased pseudoexon inclusion, which is suited to downregulate dominant negative mRNA that causes Huntington Disease.

Example 8—Further Relevant Targets

[0435] The following additional Sweet Spot sequences were identified using the criteria according to the invention (see e.g. example 2). Thus, these targets will with very high plausibility be functional targets for SSOs, allowing for incorporation of the pseudoexon in the mature mRNA.

TABLE-US-00003 TABLE 3 Relevant target genes SEQ ID NO: Gene hg38 pseudoexon coordinates Sweet Spot sequence

79	ROCK1 chr18:	21017021-21017098(-)	CTGTGAAGGCCGTGAGGTAAGAGACCTTGAC
80	ROCK1 chr18:	21022445-21022564(-)	GGAATGGGGGAAATGGAGAATAACCAGTATA
81	OGA chr10:	101795365-101795480(-)	AAGGGCGGTAGACATGAACTGAAGTCACGTT
82	OGA chr10:	101795374-101795480(-)	GGTAGATTTAAGGGCGGTAGACATGAACTGA
83	TMEM97 chr17:	28320422-28320470(+)	CGTGTAGCAAGTTGCAGTCTGGGGACTTGGT
84	TXNRD1 chr12:	104300119-104300276(+)	TGAACCCTGGGTAGCAACTCTTGAGCGAAGA
85	SLC7A11 chr4:	138191360-138191415(-)	TAGATTAGACATCAGTGGTATTGAAATTTAA
86	STAT5B chr17:	42229494-42229533(-)	TGCTCAGAGACTGGATCCTTTTAAGAGTGGG
87	MAPKAPK5 chr12:	111870683-111870782(+)	CTTATAGAGTGAAGGGTCCCTAGGCCAAGAC
88	ZYG11A chr1:	52880598-52880653(+)	TTGGAGGAGGGAGAAGCCCACCTTTTAAGGA
89	ZYG11A chr1:	52880601-52880653(+)	TTGGAGGAGGGAGAAGCCCACCTTTTAAGGA
90	MCCC2 chr5:	71642612-71642727(+)	CAGTGCCATTTAGGATGTACTGCATAAGTTT
91	MCCC2 chr5:	71642664-71642727(+)	CAGTGCCATTTAGGATGTACTGCATAAGTTT
92	SMYD2 chr1:	214334511-214334649(+)	TGGAGAGTTAGGGTTAGGATGTTAGAATTGC
93	DIAPH3 chr13:	59923001-59923041(-)	TAGCATTAACAGTATAGGTAGGGAATCTGGT
94	DIAPH3 chr13:	59923001-59923111(-)	TAGCATTAACAGTATAGGTAGGGAATCTGGT
95	COPS3 chr17:	17260072-17260112(-)	CAAGGCTACATTGGGAGACAGTGAAAGGCAG
96	SNX5 chr20:	17948072-17948142(-)	CCAGAAGCAGGATTTGCAGGGTAGGGTTATG
97	YBX1 chr1:	42699646-42699724(+)	CACTGTGCCCAACCTTGGAGTGAGATGAATA
98	CHD1L chr1:	147276877-147276929(+)	GGAGTAATAAATGTCTGTCAAGGGCAGCATC
99	PTPN11 chr12:	112456115-112456268(+)	TCAATGGATGTGCTAGCCGCTCCATTITGGCT
100	UBAP2L chr1:	154241700-154241797(+)	AATATCTGATTGTTCGGATTTATCCCAGGAAG
101	E2F3 chr6:	20456984-20457037(+)	GGCTTCAGAGATGAAGAGAGTATTTCTCCTC
102	SLC2A13 chr12:	39831850-39831885(-)	ATCAGGAAGCTTATGGTAGAAGGCTAAGGGG
103	SLC2A13 chr12:	39939133-39939264(-)	GCTATGTCAGTAGGCAACTGGAGCACAGTCC
104	ASIC1 chr12:	50079657-50079740(+)	TTGGCAGAGTTTAGCATCCAGGCAGGGTGAA
105	LRIG2 chr1:	113116745-113116798(+)	TCAGTAAAGTTTAAAGGGTAGCTAGGAGCCA
106	RNF115 chr1:	145784178-145784251(-)	CAGGCCAGGATAAGTTGTCTGAAGTCACAATG
107	HGS chr17:	81684813-81684891(+)	GTGCTTCAGGGATGAATCCAGAGGTTAACTA
108	TLK1 chr2:	171026752-171026849(-)	CTGTGTGCCCACTTAAACTGTTTTGATAATG
109	WWTR1 chr3:	149571657-149571724(-)	GAACTCAAGTAGACCCTAGTCAGCTTCAAGT
110	HMGCS1 chr5:	43297529-43297673(-)	TGTCATCCCCACTTTACAGAGAAGTGGGTTA
111	SND1 chr7:	127725671-127725726(+)	GCCTTTTCCAAACCAAAGAACAAAGCGGGAG
112	HSPG2 chr1:	21898326-21898469(-)	TTTAGGCCAAAGCAGGAATAAGATGTGGACA
113	SLC2A1 chr1:	42942082-42942153(-)	AGTGACTCCCAAAGATCAGGGTCCTGTTTGG
114	KNTC1 chr12:	122595478-122595600(+)	AGGACTATAGGCTGCGCATAGCTCGAAGGAG
115	FDFT1 chr8:	11826877-11826978(+)	AAACTCCAGGTGTCTGTAATAAGGGACAGGG
116	THOC2 chrX:	123683845-123683911(-)	

CCTGAGGTGGAGTTTGAGAGCAGCCTGAA 117 THOC2 chrX: 123706588-123706704(-) GTAGGGAGAAAAAAAAAAGCCTGTGCTGTAA 118 SMC1A chrX: 53402287-53402379(-) GATGAACACAACACAGTCTTTGCTGAAGGAG 119 PICALM chr11: 85978968-85979061(-) GGAGTTTAGAGTAGTGGATTTTATGGACTCG 120 PICALM chr11: 85984424-85984547(-) GAGGGAGAAAAAAGCTTTTATCTTGGGGCTACT 121 DYRK1A chr21: 37383084-37383160(+) AATCTGTATTCTCTAGTCAAGGAGGGGATGGG 122 PHLPP1 chr18: 62885960-62886035(+) ACTAGTAGATCACACCCCTAGATAGTTTAGC 123 AURKA chr20: 56390967-56391068(-) AATTGGAGCAAATGCCTGTAGCTTCTGTCAG 124 BACH1 chr21: 29372435-29372563(+) TTACAGTCCGTTGTAAATGAGTGACCTAGCG 125 MAP4 chr3: 47870692-47870737(-) GCCGGGGGCTCTGTAGTGGGTTTTGTGCCCA

Underlinings indicate tested or preferred binding sequence of the SSO. Binding is to the corresponding pre-mRNA. hg38 Genome coordinates follow GFF/GTF format; one-based start and end coordinates.

[0436] Further explanation of target genes:

ROCK1:

[0437] ROCK1 encodes a Rho associated serine/threonine kinase. The signaling pathway of ROCK1 has been associated with the pathogenesis of metabolic diseases and several neurodegenerative disorders, like Huntington's disease, Parkinson's disease, and Alzheimer's disease, and is a promising target for treatment of neurodegenerative disorders by suppression of its function Koch et al. 2018). Inhibition of ROCK1 is a potent target for treatment of chronic ophthalmological diseases (Moshifar et al. 2018). Hepatic ROCK1 is a suggested target for treatment of nonalcoholic fatty liver disease and hepatocellular carcinoma (Huang et al. 2018; Wu et al 2021).

[0438] Inclusion of the chr18:21022445-21022564(-) ROCK1 pseudoexon will introduce 120 bp between exon 11 and 12 in the mRNA, including an in-frame premature termination codon. mRNA transcript with inclusion of this pseudoexon is predicted as a target of nonsense-mediated decay, and increased pseudoexon inclusion will result in a reduction of gene expression.

[0439] Inclusion of the chr18:21017021-21017098(-) ROCK1 pseudoexon will introduce 78 bp between exon 12 and 13 in the mRNA. In translation of the protein, this will introduce 26 amino acids to the amino acid sequence. This will potentially reduce gene expression by disruption of protein function or alter normal protein function.

[0440] Sweet Spots for ROCK1 SSO targeting are shown in Table 3 and Table 6.

[0441] Koch J C, Tatenhorst L, Roser A E, Saal K A, Tönges L, Lingor P. ROCK inhibition in models of neurodegeneration and its potential for clinical translation. *Pharmacol Ther.* 2018 September; 189:1-21. Moshifar M, Parker L, Birdsong O C, Ronquillo Y C, Hofstedt D, Shah T J, Gomez A T, Hoopes P C S. Use of Rho kinase Inhibitors in Ophthalmology: A Review of the Literature. *Med Hypothesis Discov Innov Ophthalmol.* 2018 Fall; 7(3):101-111. Huang H, Lee S H, Sousa-Lima I, Kim S S, Hwang W M, Dagon Y, Yang W M, Cho S, Kang M C, Seo J A, Shibata M, Cho H, Belew G D, Bhin J, Desai B N, Ryu M J, Shong M, Li P, Meng H, Chung B H, Hwang D, Kim M S, Park K S, Macedo M P, White M, Jones J, Kim Y B. Rho-kinase/AMPK axis regulates hepatic lipogenesis during overnutrition. *J Clin Invest.* 2018 Dec. 3; 128(12):5335-5350. Wu H et al. (2021) *Biochemical Pharmacology Volume* 184, February 2021, 114353.

(<https://doi.org/10.1016/j.bcp.2020.114353>)

OGA

[0442] O-GlcNAc glycosylation of proteins is an important post-translational regulatory modification. The process is dynamic, and the protein O-GlcNAcase, encoded by the gene OGA, is responsible for removing the group again.

[0443] Inhibitors of OGA block cognitive decline and reduce number of amyloid plaques in animal models of Alzheimer's Disease (AD) (Yuzwa, Shan et al. 2014) and reduce amount of pathological Tau in the brain (Graham, Gray et al. 2014, Hastings, Wang et al. 2017). Furthermore, inhibition of

OGA reduces cellular internalization of α -synuclein preformed fibrils and could be a strategy for Parkinson's Disease (PD) therapy (Tavassoly, Yue et al. 2021). Insertion of the chr10:101795374-101795480(-) and chr10:101795365-101795480(-) pseudoexon located in OGA intron 10 introduces a premature stop codon targeting the resulting transcript for degradation by the NMD pathway. By functionally mimicking OGA inhibition, SSO-mediated downregulation of OGA could be a promising approach for several neuropathies, including AD and PD.

[0444] Sweet Spots for OGA SSO targeting are shown in Table 3 and Table 6. [0445] Graham, D. L., et al. 2017. *Neuropharmacology* 79: 307-313. [0446] Hastings, N. B., et al. 2017. *Mol Neurodegener* 12(1): 39. [0447] Tavassoly, O., et al. 2021. *FEBS J* 288(2): 452-470. [0448] Yuzwa, S. A., et al. 2014. *Mol Neurodegener* 9: 42.

TMEM97

[0449] Transmembrane Protein 97 (TMEM97), also known as Sigma-2 receptor, plays an important role in cholesterol homeostasis. TMEM97 has been shown to be overexpressed in several cancers, and suppression of its expression inhibits glioma cancer cell growth and metastasis (Qiu, Sun et al. 2015). TMEM97 is also involved in the pathology of neurodegenerative diseases such as Alzheimer's Disease and its inhibition may be a potential therapy (Riad, Lengyel-Zhand et al. 2020). Inhibition of TMEM97 has also been proposed as a potential therapy for Niemann-Pick type C disease (Ebrahimi-Fakhari, Wahlster et al. 2016). SSO-mediated inclusion of the chr17:28320422-28320470(+) pseudoexon in TMEM97 intron 1 introduces a premature stop codon targeting the resulting transcript for degradation by the NMD pathway, and therefore downregulates TMEM97 gene expression.

[0450] Sweet Spot for TMEM97 SSO targeting are shown in Table 3 and Table 6. [0451] Ebrahimi-Fakhari, D., et al. 2016. *Hum Mol Genet* 25(16): 3588-3599. [0452] Qiu, G., et al. 2015. *Tumour Biol* 36(10): 8231-8238. [0453] Riad, A., et al. 2020. *Mol Neurobiol* 57(9): 3803-3813.

TXNRD1

[0454] Thioredoxin Reductase 1 is encoded by TXNRD1 and is associated with unfavorable prognosis in patients with hepatocellular carcinoma (HCC) (Fu et al. 2017). In HCC tissues and cells, TXNRD1 is overexpressed and correlates positively with increasing clinical stage and shorter survival time (Fu et al. 2107). It has also been found to be mutated in several cancers, including HCC (Jia et al. 2020). It is therefore a promising therapeutic target for target down-regulation.

[0455] Transcripts including the 158nt long pseudoexon located within intron 4 of the reference transcript are subject to degradation via the NMD system due to the introduction of a frame-shift and a resulting pre-mature termination codon. It may also lead to the production of a severely truncated protein lacking the active-site amino acids necessary for reductase activity. Both scenarios result in a complete loss of function of the gene product when the transcript includes the pseudoexon.

[0456] Sweet Spot for TXNRD1 targeting are shown in Table 3 and Table 6. [0457] Fu B, et al. *Biomed Res Int*. 2017; 2017:4698167. doi: 10.1155/2017/4698167. [0458] Jia Y, et al. *Mol Clin Oncol*. 2020 December; 13(6):83. doi: 10.3892/mco.2020.2153.

SLC7A11

[0459] The solute-carrier SLC7A11 is a member of the cystine/glutamate transporter system Xc- and encodes xCT, which is overexpressed in many cancers, and is a marker of poor prognosis (reviewed in Lin et al. 2020). In glioblastoma this leads to increased glutamate secretion and neuronal death (Savaskan et al. 2008). Inhibition of xCT reduces neuronal death and edema, and prolongs survival in rats with gliomas (Savaskan et al. 2008). SLC7A11 upregulation also has an important cytoprotective effect in KRAS mutant cells by increasing intracellular antioxidant glutathione levels (Lim et al. 2019), and knock down of SLC7A11 strongly impairs growth of tumor xenografts (Lim et al. 2019). SLC7A11 is a candidate therapeutic target for both KRAS-driven tumors that are typically highly therapy-resistant, and many other cancers including gliomas. While several xCT inhibitors exist, they are less specific than an SSO mediated downregulation of

SLC7A11, and may lead to significantly more side effects when used in a clinical setting compared to an SSO based therapy.

[0460] Inclusion of the 56 nt pseudoexon located within intron 6 leads to introduction of a frame-shift and a resulting premature termination-codon, resulting in an NMD sensitive transcript, which may be down-regulated or express a truncated and non-functional protein.

[0461] Sweet Spot for SLC7A11 SSO targeting are shown in Table 3 and Table 6.

[0462] Therefore, treatment with the SSO to knock down SLC7A11 is a promising therapy for several cancers, offering higher specificity than current protein inhibitors with fewer side effects.

[0463] Lin W, et al. *Am J Cancer Res*. 2020 Oct. 1; 10(10):3106-3126. [0464] Savaskan N E, et al.

Nat Med. 2008 June; 14(6):629-32. doi: 10.1038/nm1772. [0465] Lim J K M, et al. *Mol Cell*

Oncol. 2019 Nov. 10; 7(1):1654814. doi: 10.1080/23723556.2019.1654814. [0466] Lim J K M, et

al. *Proc Natl Acad Sci USA*. 2019 May 7; 116(19):9433-9442. doi: 10.1073/pnas.1821323116.

Cancers

[0467] The PEs in the known oncogenes STAT5B (de Araujo, Erdogan et al. 2019), MCCC2 (Chen, Zhang et al. 2021), UBAP2L (Li, Wang et al. 2018), SMYD2 (Li, Zhou et al. 2018), YBX1 (Xu, Li et al. 2017), PTPN11 (Chan, Kalaitzidis et al. 2008), DIAPH3 (Rong, Gao et al. 2020), COPS3 (Zhang, Yan et al. 2018), SNX5 (Zhou, Huang et al. 2020) and ZYG11A (Wang, Sun et al. 2016) all result in inclusion of an out of frame PE. Inclusion of these PEs will therefore lead to NMD

mediated degradation of the oncogenic mRNA or production of a non-functional oncoprotein and are therefore suitable for treatment by Sweet Spot SSOs that activate inclusion SSO as cancer treatment. [0468] Chan, G., et al. 2008. *Cancer Metastasis Rev* 27(2): 179-192. [0469] Chen, et al.

2021. *Cancer Cell Int* 21(1): 22. [0470] de Araujo, E. D., et al. 2019. *Nat Commun* 10(1): 2517.

[0471] Li, L. X., et al. 2018. *Cell Death Dis* 9(3): 326. [0472] Li, Q., et al. 2018. *Med Sci Monit*

24: 7109-7118. [0473] Namour, F., et al. 2012. *Drugs R D* 12(3): 141-163. [0474] Rong, Y., et al.

2020. *J Cell Mol Med*. doi: 10.1111/jcmm.16196 [0475] Wang, X., et al. 2016. *Oncotarget* 7(7):

8029-8042. [0476] Xu, L., et al. 2018. *J Exp Clin Cancer Res* 37(1): 135. [0477] Zhou, Q., et al.

2020. *Oncogene* 39(10): 2140-2155.

Example 9—HIF1A

Aim of Study

[0478] Hypoxia inducible factor (HIF) is a transcription factor that is activated when there is a decrease in oxygen levels, or as a response to other environmental changes. HIF-1 α contributes to tumor progression in cancer by promoting signalling for angiogenesis—the formation of new blood vessels forming from already existing ones, invasiveness of the cells, metastasis and recruitment of immunosuppressive cells to the tumor environment (Tatrai et al 2014). Previous studies have shown that knock down of HIF-1 α was able to reduce tumor mass and migration of cancer cells, and activation of the oncogene is highly correlated with the risk of metastases, making HIF-1 α a possible target for anti-cancer therapy (Dai et al. 2011).

Materials and Methods

[0479] We used the analysis pipeline (FIG. 2) to analyze publicly available RNA-seq data. Panc-1 cells were grown in RPMI and transfected with 40 nM SSO using Lipofectamine. After 24 hours, cells were harvested and RNA purified. RT-PCR was performed and the resulting product visualized on 2% agarose gel. In order to determine the optimal SSOs targeting the Sweet Spot region we tested several SSOs employing 25 nt long SSOs targeting the Sweet Spot region from +9 to +13 position downstream of the 5'ss of the PE. This showed that an SSO targeting from +10 (SEQ ID NO: 128) was superior in mediating pseudoexon inclusion into the HIF1A transcript. For hypoxia experiments transfected Panc-1 or U251 cells were first incubated at normoxic condition for 48 hours, then moved to a hypoxia chamber for 24 hours. To measure cell proliferation, viability and cytotoxicity with the optimal +10 SSO, WST-1 assay was carried out on cells under normoxia and hypoxia conditions. For protein extraction cell-lysates were rapidly frozen at -80 C, to limit the time at normoxic conditions. Protein lysates were benzonase treated, and denatured

proteins were separated on a 4-12% Nupage SDS-gel, and analysed by western blotting using antibodies against HIF1 α and β -actin for control.

SSOs:

[0480] All SSO were 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety (Produced by LGC Biosearch Technologies). SSOs were used targeting position different positions inside the Sweet Spot region for that gene (relative to the 5' splice site of the pseudoexon). Thus, the SSOs binds inside the Sweet Spot.

TABLE-US-00004

TABLE	4	SEQ	ID	NO	Gene	SSO	name	SSO	sequence	5'	.fwdarw.
3'	127	HIF1A	SSO	+	9	guccuucuaaaacuaagcuugaauu	128	HIF1A	SSO	+	10
aguccuucuaaaacuaagcuugaau	129	HIF1A	SSO	+	11	caguccuucuaaaacuaagcuugaa	130	HIF1A	SSO	+	12
ucaguccuucuaaaacuaagcuuga	131	HIF1A	SSO	+	13	uucaguccuucuaaaacuaagcuug					

Results

[0481] Using the analysis pipeline (FIG. 2) to analyze publicly available RNA-seq data we identified a 34 nt pseudoexon (chr14:61724198-61724230(+)) between exon 4 and 5 in HIF1A. The pseudoexon causes a frame-shift and introduces a stop codon (UAA) in exon 5. In order to determine the optimal SSOs targeting the Sweet Spot region we tested several SSOs employing 25 nt long SSOs targeting the Sweet Spot region from +9 to +13 position downstream of the 5'ss of the PE. This showed that an SSO targeting from +10 was superior in mediating pseudoexon inclusion into the HIF1A transcript. Transfecting U251 glioblastoma cells with the +10 SSOs resulted in up to 60% inclusion of the pseudoexon (FIG. 6A). Under normoxic conditions HIF-1 α is degraded by hydroxylation, so to investigate the effect of the SSOs on a protein-level, Panc-1 cells were subjected to hypoxia, and protein extraction indicated that the +10 SSO efficiently decreased the level of HIF1A protein (FIG. 6B). WST-1 assay of U251 cells grown under hypoxic conditions had a lower viability when treated with the +10 SSO, than when treated with ctrl SSO (FIG. 6C).

Conclusion

[0482] It can be observed that it is possible to optimize the binding site within the binding region of HIF1A. The SSO performing the best was "HIF1A+10" (SEQ ID NO: 128)

Example 10—TRPM7

Aim of Study

[0483] TRPM7 belongs to the protein super family Transient Receptor Potential (TRP), which conducts the traffic of different ions across membranes. The TRP proteins work as sensors and transducers which, when activated, leads to a transmembrane flow of ions that regulate associated pathways and various physiological responses (Liu et al 2014). TRPM7 can be cleaved by caspase, splitting the kinase domain from the pore in the membrane. Studies have shown that TRPM7 is highly expressed in brain tissue and deregulation of this channel is involved in neurodegenerative diseases such as amyotrophic lateral sclerosis (ALS), parkinsonism dementia and Alzheimer's disease. It was found that TRPM7 plays a critical role in neuronal death in cases of ischemia by mediating a Ca²⁺ influx causing calcium overload resulting in oxidative stress, nitric oxide production and cell death (Leng et al 2015; Sun et al. 2009). Knock down of TRPM7 inhibit delayed neuronal cell death, which is characteristic in Alzheimer's, Huntington's, Parkinson's disease and stroke patients. This suggests that knock down of TRPM7 could be a new therapy against neuronal disorders. TRPM7 also plays a significant role in several types of cancer including glioblastoma multiforme (GBM), retinoblastoma, nasopharyngeal carcinoma, leukemia, gastric, prostate, pancreatic, breast, head and neck cancers, and it is overexpressed in pancreatic and lung cancer cells. Finally, TRPM7 plays a role in diabetes, kidney disease, and inflammatory diseases.

Materials and Methods

[0484] We used the analysis pipeline (FIG. 2) to analyze publicly available RNA-seq data. HeLa cells and U251 cells were grown in RPMI and transfected with 20 nM or nM SSO using Lipofectamine. After 24 hours, cells were harvested and RNA purified. RT-PCR was performed and

the resulting product visualized on 2% agarose gel. For Incucyte experiments U251 cells were grown in 96-well plates transfected with SSOs at 20 and 40 nM and incubated in the incucyte instrument with images taken every 4th hour. A small SSO walk was performed employing 25 nt long SSOs targeting the Sweet Spot region from +9 to +13 position downstream of the 5'ss of the PE.

SSOs:

[0485] All SSO were 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety (Produced by LGC Biosearch Technologies). SSOs were used targeting position different positions inside the Sweet Spot region for that gene (relative to the 5' splice site of the pseudoexon). Thus, the SSOs bind inside the Sweet Spot (see Table 5).

TABLE-US-00005

TABLE	5 SEQ	ID NO	Gene	SSO name	SSO sequence	5' .fwdarw.
3'	132	TRPM7 SSO	+	9	uucugacuucugcauauacuaaaga	133
		TRPM7 SSO	+	10	uuucugacuucugcauauacuaaag	134
		TRPM7 SSO	+	11	cuuucugacuucugcauauacuaaa	135
		TRPM7 SSO	+	12	acuuucugacuucugcauauacuaa	136
		TRPM7 SSO	+	13	gacuucugacuucugcauauacua	

Results

[0486] Using the analysis pipeline (FIG. 2) we identified a 60 nt pseudoexon (chr15:50588192-50588250(-)) located between exon 27 and 28 in TRPM7. The pseudoexon results in introduction of 20 amino acids in the region of the cleavage site between the channel-domain and the kinase domain of the TRPM7 protein. In order to determine the optimal SSOs targeting the Sweet Spot region we tested several 25 nt long SSOs targeting the Sweet Spot region from +9 to +13 position downstream of the 5'ss of the PE (FIGS. 5A and B). This showed that an SSO targeting from +13 was superior in mediating pseudoexon inclusion into the TRPM7 transcript. Transfection of HeLa cells or U251 glioblastoma cells with the optimal +13 targeting SSO resulted in high inclusion of the pseudoexon, which decreased growth and proliferation of cancer cells (FIG. 5C).

Conclusion

[0487] The normal expression of the TRPM7 protein is most efficiently decreased by using the SSO that binds from the +13 position and mediates a high level of pseudoexon inclusion.

REFERENCES

[0488] Tétraï E, et al. *Oncotarget* 2017, 8:44498-44510. [0489] Dai Y, et al. *International Journal of Radiation Oncology* Biology* Physics* 2011, 81:521-528. [0490] Liu M, et al. *Cell Signal* 2014, 26:2773-2781. [0491] Leng T D, et al. *CNS Neurosci Ther* 2015, 21:252-261. [0492] Sun H S, et al. *Nat Neurosci* 2009, 12:1300-1307.

Example 11—Further Relevant Targets Experimentally Validated

[0493] The following Sweet Spot sequences were identified in disease associated genes using the criteria according to the invention (see e.g. example 2) and demonstrated by functional testing to be targets for SSOs, allowing for incorporation of the pseudoexon in the mature mRNA.

Materials and Methods

[0494] We used public RNA-sequencing data (Geuvadis, E-MTAB-2836, E-MTAB-513, GSE52946, and GSE124439) and mapped them with STAR after trimming for adapter contamination and poor quality bases with bbduk. HeLa cells were seeded in 12-well plates and forward transfected at 60% confluence with 20 or 40 nM 2'-O-methyl SSOs with full phosphorothioate backbone using Lipofectamine RNAiMAX (invitrogen). A non-binding ctrl SSO (5'GCUCAUAUAUGCUACUGCCAUGCUUG3') (SEQ ID NO: 126) was used as control. RNA was harvested after 48 hours using Trizol (Invitrogen) and chloroform to isolate the RNA, followed by precipitation with isopropanol. Complementary DNA (cDNA) was synthesized from 500 ng RNA using the High capacity cDNA kit (Applied Biosystems). Primers were designed to span at least one exon-exon junction of the neighboring exons flanking the pseudoexons of interest. PCR was carried out using TEMPase Hot Start DNA polymerase (ampliqon) and 1 µl cDNA per reaction. 0.5 pmol/µl of each primer was used. The PCR products were separated on a 2% Seakem

LE (Lonza) TBE agarose gel, for 1 hour at 80V.

[0495] The Sweet Spot region is located +9 to +39 of the 5' splice site of the pseudoexon.

[0496] All SSOs were 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety (Produced by LGC Biosearch Technologies). SSOs were used targeting position +11 to +35 inside the Sweet Spot region for that gene (relative to the 5' splice site of the pseudoexon). Thus, the SSOs bind inside the Sweet Spot.

Results

[0497] Using our double-junction approach, we identified further 46 fully spliced pseudoexons and tested SSOs targeting the Sweet Spot region from +11 to +35 downstream of the selected pseudoexons identified in RNA sequencing data (table 6). All 46 of these selected SSOs were able to increase pseudoexon inclusion into the mRNA transcript of the targeted gene (Table 6 below).

[0498] Table 6: 46 pseudoexons matching all criteria, all activated by an SSO located within the Sweet Spot region. Sweet Spot region is annotated by its genomic sequence (DNA).

TABLE-US-00006 TABLE 6 Targets functionally validated as responsive to SSO targeting the Sweet Spot sequence. SSO hg38 seq pseudoexon ID Gene coordinates Sweet Spot seq (+9 to +39) 79* ROCK1 chr18: 21017021-

CTGTGAAGGCCGTGAGGTAAGAGACCTT 21017098(-) GAC 80* ROCK1 chr18: 21022445- GGAATGGGGGAAATGGAGAATAACCAGT 21022564(-) ATA 81* OGA chr10: 101795365- AAGGGCGGTAGACATGAACTGAAGTCAC 101795480(-) GTT 82* OGA chr10: 101795374- GGTAGATTTAAGGGCGGTAGACATGAAC 101795480(-) TGA 83* TMEM97 chr17: 28320422- CGTGTAGCAAGTTGCAGTCTGGGGACTT 28320470(+) GGT 85* SLC7A11 chr4: 138191360- TAGATTAGACATCAGTGGTATTGAAATTT 138191415(-) AA 87* MAPKAPK5 chr12: 111870683- CTTATAGAGTGAAGGGTCCCTAGGCCAA 111870782(+) GAC 90* MCCC2 chr5: 71642612- CAGTGCCATTTAGGATGTACTGCATAAGT 71642727(+) TT 91* MCCC2 chr5: 71642664- CAGTGCCATTTAGGATGTACTGCATAAGT 71642727(+) TT 93* DIAPH3 chr13: 59923001- TAGCATTAAACAGTATAGGTAGGGAATCTG 59923041(-) GT 94* DIAPH3 chr13: 59923001- TAGCATTAAACAGTATAGGTAGGGAATCTG 59923111(-) GT 99* PTPN11 chr12: 112456115- TCAATGGATGTGCTAGCCGCTCCATTG 112456268(+) CT 100* UBAP2L chr1: 154241700- AATATCTGATTGTCGGATTTATCCCAGGA 154241797(+) AG 105* LRIG2 chr1: 113116745- TCAGTAAAGTTTAAAGGGTAGCTAGGAGC 113116798(+) CA 106* RNF115 chr1: 145784178- CAGGCCAGGATAAGTTGTGCGAAGTCACA 145784251(-) ATG 107* HGS chr17: 81684813- GTGCTTCAGGGATGAATCCAGAGGTTAAC 81684891(+) TA 110* HMGS1 chr5: 43297529- TGTCATCCCCACTTTACAGAGAAGTGGGT 43297673(-) TA 111* SND1 chr7: 127725671- GCCTTTTCCAAACCAAAGAACAAGCGGG 127725726(+) AG 114* KNTC1 chr12: 122595478- AGGACTATAGGCTGCGCATAGCTCGAAG 122595600(+) GAG 117* THOC2 chrX: 123706588- GTAGGGAGAAAAAAGCCTGTGCTG 123706704(-) TAA 118* SMC1A chrX: 53402287- GATGAACACAACACAGTCTTTGCTGAAGG 53402379(-) AG 122* PHLPP1 chr18: 62885960- ACTAGTAGATCACACCCCTAGATAGTTTA 62886035(+) GC 123* AURKA chr20: 56390967- AATTGGAGCAAATGCCTGTAGCTTCTGTC 56391068(-) AG 137 ATXN7 chr3: 63921276- TTAAACCAGTGGTCCCCAGGCTTTATATT 63921329(+) C 138 DDR2 chr1: 162666267- TTGTGGAATGAATGAATGAGCAAATGAAG 162666368(+) GA 139 HIF1A chr14: 61724148- AATTCAAGCTTAGTTTATGAAGGACTGAA 61724230(+) CA 140 HTT chr4: 3223670- GAAGCTGGAAACGTGACAGGAAGTACG 3223729(+) TGG 141 LRRK2 chr12: 40362438- AAAATACATTGTCCTCATCCTTATGAAATT 40362491(+) A 142 LRRK2 chr12: 40362410- AAAATACATTGTCCTCATCCTTATGAAATT 40362491(+) A 143 RAP1GDS1 chr4: 98419239- TACTGCACCAGGCCTTAGGCATCTTTAGT 98419331(+) TG 144 TTBK2 chr15: 42809105- TGGGTGTGTTCTAATGACTAGTATCAATG 42809208(-)

TG 145 TTK chr6: 80034193- CCATAAAACTAACAATCACAACTCTGCAC 80034275(+) AC
 146 BUD31 chr7: 99419162- CTATATGGCATGGTGGCAGGTCCTTCGTG 99419275(+) GG
 147 CD44 chr11: 35188146- TTGTCCCTGACCTGTGCCTTGCAATAGGA 35188238(+) AG
 148 CDKL5 chrX: 18623879- TTTTATGTATGGTAGCCCTGAAAAC TGCT 18624001(+) CC
 149 ITGBL1 chr13: 101705140- GTGGTATAACCCAGGGAAAATCTCTCTTA 101705240(+) TT
 150 LRP6 chr12: 12149294- AAGAAATACTCCTTAAGACTTGAGAGAGC 12149360(-) CT
 151 MELK chr9: 36654164- AGAAAGCTATTAGATACGTATACCTCTAT 36654224(+) GC
 152 PDE4D chr5: 59036523- TGCCACACCTTCAGTCTGAACTGGAAATG 59036587(-) TG
 153 PMPCB chr7: 103305205- AGAAGGTTTCCTTGAGGTCATGTTTGAAG 103305273(+) CC
 154 RBPJ chr4: 26388182- CACGGTGCCCAGCCTAAAACAGTTATTAT 26388277(+) AT
 155 ROCK2 chr2: 11198121- CTAGATACTCTTCACTGTTGTTGCACATA 11198215(-) GA
 156 TASP1 chr20: 13415123- GCATCTTAAAATATTTAAAATTTAAAAACT 13415262(-) A
 157 ATP5F1C chr10: 7805451- ATTGAGACCATCCTGGCTAACTCGGTGAA 7805532(+) AC
 158 LRRK2 chr12: 40362438- AAAATACATTGTCCTCATCTTTATGAAATT 40362491(+) A
 159 LRRK2 chr12: 40362410- AAAATACATTGTCCTCATCTTTATGAAATT 40362491(+) A
 *Also included in table 3. Underlining indicates tested binding sequence of the SSO. Binding is to the corresponding pre-mRNA. hg38 Genome coordinates follow GFF/GTF format; one-based start and end coordinates.

[0499] In Table 6, the Sweet Spots for SEQ ID 141 and 142 correspond to the major allele (C) of rs17444202. The Sweet Spots for SEQ ID 158 and 159 correspond to the minor allele (T) of rs17444202.

[0500] Seq ID NO's: 160-180 in Table 7 are further pseudoexons in genes where at least one other pseudoexon has already been activated by a SSO targeting the Sweet Spot (genes listed in table 1 and table 6). Seq ID NO's: 181-201 in Table 7 are additional Sweet Spot sequences identified using the criteria according to the invention (see e.g. example 2). Thus, these targets will with very high plausibility be functional targets for SSOs, allowing for incorporation of the pseudoexon in the mature mRNA.

TABLE-US-00007 TABLE 7 relevant target pseudoexons. SSO seq hg38 ID Gene
 coordinates Sweet Spot seq 160 RNF115 chr1: 145794074-
TTCGAGACCAGCCTGGCCATCGTGGCGAA 145794196(-) AC 161 RNF115 chr1:
 145784113- TTCCAGGGCTTACCTCTAGTCAACACAGTA 145784251(-) G 162 RNF115
 chr1: 145780773- AAGAGGGCAGGGTGCTCTGAGCTGATTTA 145780881(-) TA 163
 RNF115 chr1: 145759389- GGAGGAAAATATTTAAGAGTTCATTCAGTT 145759535(-) T
 164 LRP6 chr12: 12184211- TTTGGCTTATCACAGTTGTTTCATGTTATGG 12184244(-) T 165
 LRP6 chr12: 12212260- AGACCATTATTAAATATTTGAGTGCTGACT 12212305(-) G 166
 LRP6 chr12: 12197784- CACCATGCCTGGCCATAAATTTAATTCTTA 12197878(-) T 167
 UBAP2L chr1: 154249958- TCACTTAATGTCATTGATAGGTTCTTGCGA 154250030(+) C
 168 UBAP2L chr1: 154249997- TCACTTAATGTCATTGATAGGTTCTTGCGA 154250030(+) C
 169 UBAP2L chr1: 154222819- CAAGATTCTTCATCTCCAGTGTGATAGCTG
 154222986(+) G 170 UBAP2L chr1: 154241738-
AATATCTGATTGTCGGATTATCCCAGGAA 154241797(+) G 171 UBAP2L chr1:
 154262173- GACCATACTTAATAGAGTGGAATTGGAC 154262202(+) TG 172 ZNF558
 chr19: 8831564- AGTCATGGCCCACAGGCTGGGCCCTTATT 8831582(-) AC 173 ZNF558
 chr19: 8831564- AGTCATGGCCCACAGGCTGGGCCCTTATT 8831647(-) AC 174 ZNF558
 chr19: 8831564- AGTCATGGCCCACAGGCTGGGCCCTTATT 8831642(-) AC 175 KNTC1
 chr12: 122585408- ATGAGGCCACTTCTCTGTTCCAGTCACTTG 122585420(+) A 176
 KNTC1 chr12: 122564657- CCCAGTGCTCATTTCTTTAACACAAATATT 122564756(+) T
 177 KNTC1 chr12: 122621236 ACCAGTAGATAGAGATCTACATAAATGAAC -122621288(+) A
 178 KNTC1 chr12: 122620698 GAAAGATAGATTAAGGTCCAGCTGAGGGC
 -122620759(+) TG 179 KNTC1 chr12: 122554273

CACCATACCCAGCAAGAATAATCTGTGTA -122554359(+) T 180 KNTC1 chr12:
 122528435 AACTAAATTAGTCCCTTGGTATTGGTATCC -122528583(+) A 181 SMC1A
 chrX: 53404716- TTGGAGACCAGCCTGGCCAAACCTGTCTC 53404804(-) TA 182 SMC1A
 chrX: 53407427- GAAGAGATGTGAAGCTTCCATGCCCAATC 53407474(-) TG 183 SMC1A
 chrX: 53407371- ACACAGGGGCTTCATTATGTAGTCATGATT 53407474(-) G 184 FLT1
 chr13: 28337518- CTCTGATTGATTAACGTGGGAGCAAGGAG 28337637(-) GC 185 FLT1
 chr13: 28414636- CTCTCGATGAAATCAGATAATAAACTGACA 28414760(-) A 186 FLT1
 chr13: 28435084- TGAGTTAATGTTATGAATCTTGGAGGACCT 28435140(-) G 187 FLT1
 chr13: 28435084- TGAGTTAATGTTATGAATCTTGGAGGACCT 28435205(-) G 188
 EFEMP1 chr2: 55870349- CACTAATCATTGATGGTTAATTAATTATAC 55870498(-) A 189
 CLCN1 chr7: 143319065- TAAAGGAAAGAGGTTTAATGGGCCCACAG 143319127(+) TT
 190 CD5L chr1: 157836389- CTAATTATCTAGCTGTGGGAAGAAGTAGA 157836492(-) TG
 191 A2M chr12: 9072149- AAGTAGAGTTTCTTGACAGGGTACCATTAA 9072182(-) G 192
 MAOB chrX: 43778486- TTCCCTCAATAACAATAAGTAAATATTAC 43778558(-) C 193
 MAOB chrX: 43837770- ACTTGCAATATGCTGTGGTTTAGGTGATGC 43837923(-) T 194
 HSP90AA1 chr14: 102117585 CACAGGAACCTTATTCCTCACTGGTAGAAAA -102117727(-)
 C 195 ALK chr2: 29253737- TTTCTGCAGCTGGGGGGTGTTCAGCTCCT 29253814(-) AG
 196 JPH3 chr16: 87695507- AGCCTGGGTCTGCAGAACAGGTTTCAAGG 87695657(+) AA
 197 HNF1B chr17: 37711993- TCTGTAGGGCAGCACCCACCCCTGCCGTC 37712084(-) CT
 198 P2RX7 chr12: 121149278- TAAAGGAAAGAGGTTTAATGGGCCCACAG 121149369(+)
 TT 199 P2RX7 chr12: 121175867- CAGCACCTTCGATCTGACCAGTTTAGCAAC
 121175980(+) C 200 DBI chr2: 119369029- TACAGTGAGTTCGTAAACTCTGTCTTCCA
 119369176(+) G 201 DBI chr2: 119371982- GATACTGTCTCCTGTAATTAGTAGAATCTC
 119372085(+) A Underlining indicates tested binding sequence of the SSO. Binding is to the
 corresponding pre-mRNA. hg38 Genome coordinates follow GFF/GTF format; one-based start and
 end coordinates.

TABLE-US-00008 TABLE 8 SSOs targeting SMAD2 and LRRK2 SEQ ID NO Gene											
SSO	name	SSO	sequence	5'	.fwdarw.	3'	202	SMAD2	SMAD2SSO	+	9
UCUCGCUUAGCUUCAACAACUAAGAC	203	SMAD2	SMAD2SSO	+	10						
AUCUCGCUUAGCUUCAACAACUAAGA	204	SMAD2	SMAD2SSO	+	11						
CAUCUCGCUUAGCUUCAACAACUAAG	205	SMAD2	SMAD2SSO	+	12						
GCAUCUCGCUUAGCUUCAACAACUAA	206	SMAD2	SMAD2SSO	+	13						
UGCAUCUCGCUUAGCUUCAACAACUA	207	SMAD2	SMAD2SSO	+	14						
CUGCAUCUCGCUUAGCUUCAACAACU	208	LRRK2	LRRK2SSO	+	11						
UUCAUAAAGACGAGGACAAUGUAUU	209	LRRK2	LRRK11CSSOC								
UUCAUAAGGACGAGGACAAUGUAUU	210	LRRK2	LRRK11CSSOT								
UUCAUAAAGACGAGGACAAUGUAUU	211	LRRK2	LRRK9CSSOC								
CAUAAGGACGAGGACAAUGUAUUUU	212	LRRK2	LRRK9CSSOT								
CAUAAAGACGAGGACAAUGUAUUUU	213	LRRK2	LRRK13CSSOC								
AUUUCAUAAGGACGAGGACAAUGUA	214	LRRK2	LRRK13CSSOT								
AUUUCAUAAAGACGAGGACAAUGUA	215	LRRK2	LRRK11SSOC								
UUCAUAAGGAUGAGGACAAUGUAUU	216	LRRK2	LRRK11SSOT								
UUCAUAAAGAUGAGGACAAUGUAUU											

Example 12—RNF115

[0501] RNF115 (Ring Finger Protein 115), previously named Breast Cancer Associated 2 (BCA2), is a RING-finger E3 ubiquitin that mediates polyubiquitination of substrates. RNF115 causes ubiquitination and proteasomal degradation of the tumor suppressor p21 in breast cancer (Wang et al. 2013). In lung cancer RNF115 also functions as an oncogene by regulating Wnt/ β -catenin pathway via ubiquitination of adenomatous polyposis coli (APC) leading to increased proliferation (Wu et al. 2021).

[0502] RNF115 is associated with breast cancer. It is overexpressed in more than 50% of invasive breast cancers, and its up-regulation correlates with estrogen receptor positive (ER+) status and poor prognosis.

[0503] Therefore, inhibiting RNF115 activity will inhibit the β -catenin and function to inhibit cancers, such as lung cancer and breast cancer.

[0504] Through in silico analysis and in vivo experiments, we aimed at investigating the presence of pseudoexons in the RNF115 gene, which may be used to down-regulate the expression of this gene.

Materials and Methods

[0505] We used our novel double-junction approach to identify fragments with fully spliced pseudoexons in publicly available RNA-seq data from the GEUVADIS consortium, E-MTAB-2836, E-MTAB-513, GSE52946, and GSE124439. HeLa cells, NCI-H23 lung cancer cells and NCI-H23 lung cancer cells were seeded in 12-well plates and forward transfected at 60% confluence with 5 nM, 10 nM or 20 nM SSO using Lipofectamine RNAiMAX (Invitrogen).

[0506] SSO (targeting SEQ ID NO: 106; specific target sequence is underlined in SEQ ID NO: 106 in table 3 and table 6) was a 25 nt long phosphorothioate RNA oligonucleotides with 2'-O-methyl modification on each sugar moiety (Produced by LGC Biosearch Technologies). The SSO is complementary to position +11 to +35 inside the Sweet Spot region for the RNF115 gene (relative to the 5' splice site of the pseudoexon). Thus, the SSOs binds inside the Sweet Spot.

[0507] A non-binding SSO (5'GCUCAAUAUGCUACUGCCAUGCUUG3') (SEQ ID NO: 126) with similar modifications was used as a negative control. RNA was harvested after 48 hours using Trizol (Invitrogen) and chloroform to isolate the RNA, followed by precipitation with isopropanol. Complementary DNA (cDNA) was synthesized from 500 ng RNA using the High capacity cDNA kit (Applied Biosystems). Primers were designed to span at least one exon-exon junction of the neighboring exons flanking the pseudoexons of interest. PCR was carried out using TEMPase Hot Start DNA polymerase (ampliqon) and 1 μ l cDNA per reaction. 0.5 pmol/ μ l of each primer was used. The PCR products were separated on a 2% Seakem LE (Lonza) TBE agarose gel, for 1 hour at 80V. Protein was harvested after 72 hours for western blotting and denatured proteins were separated on a 4-12% NuPAGE SDS-gel. Western blot analysis was performed with antibodies directed towards RNF115 (ab187642, Abcam) β -catenin (#9587, Cell Signaling Technology), β -actin for control (ab8229, Abcam).

[0508] To measure cell proliferation, viability and cytotoxicity with the optimal +11 SSO, NCI-H23 cells were reverse transfected with SSOs in a concentration gradient, and WST-1 assay (Roche) was carried out on cells 48 hours after transfection.

Results

[0509] Using our double-junction approach to examine fragments with fully spliced pseudoexons, we identified a pseudoexon within intron 3 of the RNF115 gene. Inclusion of the chr1:145784178-145784251 RNF115 pseudoexon introduces 74 bp between exon 3 and 4 in the mRNA. This will lead to a reading frame-shift and a resulting pre-mature termination codon leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein (of only 84 amino acids) lacking the functional domains. This truncated protein could function as a decoy by binding to substrates and inhibit polyubiquitination and function to inhibit cancers (Table 3 and Table 6).

[0510] Transfection of NCI-H23 cancer cells with an SSO complementary to position +11 to +35 downstream of the 5' splice site (SEQ ID NO: 106) resulted in up to 90% more inclusion of the pseudoexon (FIG. 7A).

[0511] Growth of NCI-H23 cancer cells is inhibited by treatment by SSO-mediated down-regulation as shown by WST-1 assay (FIG. 7B). Furthermore, RNF115 protein levels are severely decreased following treatment of NCI-H23 lung cancer cells as shown by western blotting (FIG. 7C). The decreased levels of functional RNF115 caused decreased levels of β -catenin as shown by western blotting (FIG. 7C).

Conclusion

[0512] The normal expression of the RNF115 gene product is decreased by at least 90% using a specific SSO to increase inclusion of a pseudoexon, which disrupts the function of the normal gene product. This reduces β -actin expression and growth of lung cancer cells showing SSO based activation of the RNF115 pseudoexon works to inhibit cancer.

[0513] Similarly, inclusion of the chr1: 145794075-145794196 RNF115 pseudoexon introduces 122 bp between exon 1 and 2 in the mRNA. This will lead to a reading frame-shift and a resulting pre-mature termination codon leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein lacking the functional domains. This truncated protein could function as a decoy by binding to substrates and inhibit polyubiquitination and function to inhibit cancers (Table 7).

[0514] Similarly, inclusion of the chr1: 145784114-145784251 RNF115 pseudoexon introduces 138 bp between exon 3 and 4 in the mRNA. This will lead to a premature stop codon only twelve codons downstream of glutamic acid 73 leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein (of only 84 amino acids) lacking the functional domains. This truncated protein could function as a decoy by binding to substrates and inhibit polyubiquitination and function to inhibit cancers (Table 7).

[0515] Similarly, inclusion of the chr1: 145780774-145780881 RNF115 pseudoexon introduces 108 bp between exon 3 and 4 in the mRNA. This will lead to a premature stop codon only twenty-four codons downstream of glutamic acid 73 leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein (of only 96 amino acids) lacking the functional domains. This truncated protein could function as a decoy by binding to substrates and inhibit polyubiquitination and function to inhibit cancers (Table 7).

[0516] Similarly, inclusion of the chr1: 145759390-145759535 RNF115 pseudoexon introduces 146 bp between exon 4 and 5 in the mRNA. This will lead to a reading frame-shift after glycine 143 and a resulting pre-mature termination codon 94 codons downstream leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein (of 226 amino acids) lacking the functional domains. This truncated protein could function as a decoy by binding to substrates and inhibit polyubiquitination and function to inhibit cancers (Table 7). [0517] Wang Z, Nie Z, Chen W, Zhou Z, Kong Q, Seth A K, et al. (2013) RNF115/BCA2 E3 ubiquitin ligase promotes breast cancer cell proliferation through targeting p21Waf1/Cip1 for ubiquitin-mediated degradation. *Neoplasia*. 15(9):1028-35. [0518] Wu X T, Wang Y H, Cai X Y, Dong Y, Cui Q, Zhou Y N, Yang X W, Lu W F, Zhang M. (2021) RNF115 promotes lung adenocarcinoma through Wnt/ β -catenin pathway activation by mediating APC ubiquitination. *Cancer Metab*. 9(1):7. Example 13—LRRK2

Aim of Study

[0519] Parkinson's disease is a progressive neurodegenerative disorder characterized by loss of dopaminergic neurons that affects movement control. Mutations in the leucine-rich repeat kinase 2 (LRRK2) gene account for common risk factors associated with Parkinson's disease (Alessi & Sammler, 2018). Dominantly inherited and sporadic pathogenic mutations in LRRK2 causes hyperactivation of the LRRK2 kinase, and downregulation of LRRK2 gene expression is a potential treatment strategy.

Materials and Methods

[0520] We used the analysis pipeline (FIG. 2) to identify fragments with fully spliced pseudoexons in publicly available RNA-seq data. HeLa and U251 cells were reverse transfected with 20 nM SSO (targeting SEQ ID NO: 141 and SEQ ID NO: 142; specific target sequence is underlined in SEQ ID NO: 141 and SEQ ID NO: 142 in table 3b) using Lipofectamine RNAiMAX (invitrogen). A non-targeting SSO (5'-GCUCAAUAUGCUACUGCCAUGCUUG-3') (SEQ ID NO: 126) was used as a negative control. Cells were harvested after 48 hours using Trizol (Invitrogen) and RNA was extracted using chloroform and isopropanol. Complementary DNA (cDNA) was synthesized

using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems) and PCR was carried out using the TEMPase Hot Start DNA Polymerase (Ampliqon) with primers in LRRK2 exon 47 and 48. The PCR products were separated on a 1.5% SeaKem LE (Lonza) TBE agarose gel.

Results

[0521] Using our analysis pipeline (FIG. 2), we identified two LRRK2 pseudoexons that can be activated by an SSO targeting the Sweet Spot. The two pseudoexons; chr12:40362438-40362491(+) and chr12:40362410-40362491(+), are located within intron 47 and have the same 5' splice site. Transfection of HeLa and U251 cells with an SSO complementary to position +11 to +35 downstream of the 5' splice site (SEQ ID NO: 215) resulted in increased inclusion of both pseudoexons between exon 47 and 48 in the LRRK2 mRNA transcript (FIG. 9). The chr12:40362438-40362491(+) LRRK2 pseudoexon is 54 nt long and pseudoexon inclusion will introduce 18 amino acids to the WD40 domain of the translated LRRK2 protein. The chr12:40362410-40362491(+) LRRK2 pseudoexon is 82 nt long and pseudoexon inclusion will cause a frame-shift and insertion of a premature termination codon, and target the transcript for degradation by nonsense-mediated mRNA decay.

[0522] It was surprisingly discovered that the identified Sweet Spot sequence harbors a SNP, rs17444202. We therefore used SSOs targeting either SEQ ID NO: 141 and 142 corresponding to the major allele (C) of rs17444202 or SSOs targeting SEQ ID 158 and 159 corresponding to the minor allele (T) of rs17444202 (see table 6 in example 11). Further, we used SSOs with additional mismatches to achieve preferential targeting of either the major or minor allele. All SSOs induced inclusion of the chr12:40362438-40362491(+) LRRK2 pseudoexon and the chr12:40362410-40362491(+) LRRK2 pseudoexon (results not shown). The sequences of the employed SSOs are listed in table 8 in example 11, with SEQ ID NO's: 200-207.

Conclusion

[0523] The normal and hyper-activated function of LRRK2 will be decreased by using a specific SSO targeting the Sweet Spot to induce pseudoexon inclusion and thereby reduce the expression and activity of the normal LRRK2 gene product.

[0524] Pseudoexon inclusion that introduces amino acids to the translated sequence will potentially reduce gene expression by disruption of protein function or alter normal protein function. Pseudoexon inclusion that causes frame-shift with insertion of a premature termination codon will reduce gene expression by degradation of the transcript or translation of a truncated and non-functional protein.

[0525] Without being bound by theory, by the identification of a SNP in the Sweet Spot region it is possible to make allele-specific targeting (or at least allele-preferred targeting), by screening the subjects SNP status before selecting SSOs and initiating a treatment. [0526] Reference: Alessi, D. R. & Sammler E (2018) Science, 360(6384):36-37. DOI: 10.1126/science.aar5683

Example 14—LRP6

[0527] LRP6 (LDL Receptor Related Protein 6) is a member of the low density lipoprotein (LDL) receptor gene family. LRP6 functions as a receptor and co-receptor for Wnt in the Wnt/beta-catenin signaling cascade and plays a role in the regulation of cell differentiation, proliferation, and migration. It is also involved in glucose and lipid metabolism signaling. Inhibition of LRP6 can be a therapeutic option for cancers, such as breast-, liver- and colorectal-cancer, as well as metabolic and neurodegenerative disease (Reviewed by Jeong and Jho 2021).

[0528] Inclusion of the chr12:12149294-12149360(−) LRP6 pseudoexon introduces 67 bp between exon 13 and 14 in the mRNA. This will lead to a reading frame-shift and a resulting pre-mature termination codon leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein (of only 1006 amino acids) lacking the transmembrane and cytosolic domains. This truncated protein could function as a decoy receptor for Wnt proteins and thereby inhibit Wnt signaling and function to inhibit cancers and other diseases. (Table 6—Functionally

validated)

[0529] Similarly, inclusion of chr12:12184211-12184244(-) LRP6 pseudoexon will introduce 34 bp between exon 4 and 5 in the mRNA. This will lead to a reading frame-shift and a resulting premature termination codon leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein lacking the transmembrane and cytosolic domains. This truncated protein could function as a decoy receptor for Wnt proteins and thereby inhibit Wnt signaling and function to inhibit cancers and other diseases (Table 7).

[0530] Similarly, inclusion of chr12:12212260-12212305(-) LRP6 pseudoexon will introduce 46 bp between exon 1 and 2 in the mRNA. This will lead to a reading frame-shift and a resulting premature termination codon leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein lacking the transmembrane and cytosolic domains. This truncated protein could function as a decoy receptor for Wnt proteins and thereby inhibit Wnt signaling and function to inhibit cancers and other diseases (Table 7).

[0531] Similarly, inclusion of chr12:12197784-12197878(-) LRP6 pseudoexon will introduce 95 bp between exon 3 and 4 in the mRNA. This will lead to a reading frame-shift and a resulting premature termination codon leading to degradation via the NMD system. It will also lead to the production of a severely truncated protein lacking the transmembrane and cytosolic domains. This truncated protein could function as a decoy receptor for Wnt proteins and thereby inhibit Wnt signaling and function to inhibit cancers and other diseases (Table 7).

Claims

1.-23. (canceled)

24. A composition comprising a splice switching oligonucleotide (SSO), said composition comprising: an SSO complementary or substantially complementary to region within a nucleic acid selected from the group consisting of: a nucleic acid according to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or an SSO selected from the group consisting of: a nucleic acid according to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; wherein said SSO being complementary or substantially complementary to a target pre-mRNA, said target pre-mRNA comprising: a function-disabling pseudoexon comprising: at the 5'-end a 3' splice site; and at the 3'-end a 5' splice site; wherein said SSO is complementary or substantially complementary to the target pre-mRNA at a region +9 to +39 downstream to the 5' splice site of said pseudoexon; and wherein, when said SSO, in vivo, hybridizes to the pre-mRNA within the region +9 to +39 downstream to the 5' splice site of said pseudoexon; said pseudoexon becomes part of the mature mRNA to a greater extent compared to corresponding pre-mRNA not contacted with the SSO.

25. The composition according to claim 24, wherein the SSO is complementary or substantially complementary to region within a nucleic acid selected from the group consisting of a nucleic acid according to any of SEQ ID NO: 106, SEQ ID NO: 21 and any of SEQ ID NO's: 141, 142, 158 and 159; a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 106, 21, 141, 142, 158 and 159; or a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 106, 21, 141, 142, 158 and 159.

26. The composition according to claim 24, wherein said SSO comprises a sequence, which is substantially complementary to the polynucleotide in the pre-mRNA, comprises at the most 3 mismatches.

- 27.** The composition according to claim 24, wherein said SSO comprises a sequence which is complementary or substantially complementary to a polynucleotide in the pre-mRNA as defined in claim 1, wherein said sequence has a length in the range 9-31 nucleotides.
- 28.** The composition for use according to claim 24, wherein said SSO comprises one or more artificial nucleotides.
- 29.** The composition for use according to claim 24, wherein the oligonucleotide does not mediate RNase H mediated degradation of the mRNA.
- 30.** A method for treating or alleviating a disease in a subject, the method comprising administering to a subject in need thereof, a composition comprising a splice switching oligonucleotide (SSO), said composition comprising: an SSO complementary or substantially complementary to region within a nucleic acid selected from the group consisting of: a nucleic acid according to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 106, 1-26, 79-105, 107-125 and 137-201; or an SSO selected from the group consisting of: a nucleic acid according to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; or a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 127-136 and SEQ ID NO's: 202-216; wherein said SSO being complementary or substantially complementary to a target pre-mRNA, said target pre-mRNA comprising: a function-disabling pseudoexon comprising: at the 5'-end a 3' splice site; and at the 3'-end a 5' splice site; wherein said SSO is complementary or substantially complementary to the target pre-RNA at a region +9 to +39 downstream to the 5' splice site of said pseudoexon; and wherein, when said SSO, in vivo, hybridizes to the pre-mRNA within the region +9 to +39 downstream to the 5' splice site of said pseudoexon; said pseudoexon becomes part of the mature mRNA to a greater extent compared to corresponding pre-mRNA not contacted with the SSO.
- 31.** The method according to claim 30, wherein the subject has a disease or condition characterized by increased expression or altered function of the disorder-causing or disorder-influencing functional protein, or where decreased expression of the functional gene product is therapeutically beneficial.
- 32.** The method according to claim 30, wherein the disease is selected from the group consisting of: cancer, Inflammatory diseases, Neurodegenerative or neurological diseases, Metabolic conditions, Chronic liver disease and Inherited retinal dystrophies (IRDs).
- 33.** The method according to claim 30, wherein the disease is cancer and the SSO is complementary or substantially complementary to the pre-RNA selected from the group consisting of: ROCK1, LRP6, TXNRD1, SLC7A11, STAT5B, MAPKAPK5, ZYG11A, MCCC2, SMYD2, DIAPH3, COPS3, SNX5, YBX1, CHD1L, PTPN11, UBAP2L, RNF115, HGS, TLK1, WWTR1, HMGCS1, SND1, THOC2, E2F3, LRIG2, HSPG2, SLC2A1, KNTC1, FDFT1, SMC1A, HIF1A, CSPP1, TRPM7, DDR2, STAG2, ORC1 and TAF2.
- 34.** The method according to claim 30, wherein the disease is an inflammatory disease and the SSO is complementary or substantially complementary to the pre-RNA selected from the group consisting of DDR2 and TRPM7.
- 35.** The method according to claim 30, wherein the disease is a neurodegenerative or neurological disease and the SSO is complementary or substantially complementary to the pre-RNA selected from the group consisting of LRRK2, ROCK1, OGA, TMEM97, PICALM, E2F3, SLC2A13, ASIC1, TRPM7 and LRIG2.
- 36.** The method according to claim 30, wherein the disease is Chronic liver disease and the SSO is complementary or substantially complementary to the pre-RNA selected from the group consisting of SMAD2 and TRPM7.
- 37.** The method according to claim 30, wherein the disease is diabetes and the SSO is

complementary or substantially complementary to the pre-RNA selected from the group consisting of TXNRD1, DYRK1A, TRPM7 and PHLPP1.

38. The method according to claim 30, wherein the composition is administered to a subject who is heterozygous in the pre-mRNA region targeted by the SSO, resulting in the SSO having an increased binding affinity to pre-mRNA of one of the alleles, to provide an increased splice switching activity in said allele.

39. The method according to claim 30, wherein the SSO promotes inclusion of a pseudo-exon to a greater extent of a disease-causing allele compared to the other allele.

40. The method according to claim 30, wherein the disease is Parkinson's disease, the pre-mRNA encodes for LRRK2, and the subject is heterozygous at the rs17444202 position.

41. The method according to claim 30, wherein the disease is Parkinson's disease, the pre-mRNA encodes for LRRK2, and the SSO is complementary or substantially complementary to a region within a nucleic acid selected from the group consisting of a nucleic acid according to any of SEQ ID NO's: 141, 142, 158 and 159; or a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 141, 142, 158 and 159; or a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 141, 142, 158 and 159; or the SSO is selected from the group consisting of: a nucleic acid according to any of SEQ ID NO's: 208-216; or a nucleic acid comprising 1 or 2 or 3 substitutions when compared to any of SEQ ID NO's: 208-216; or a nucleic acid sequence having at least 90% sequence identity to any of SEQ ID NO's: 208-216.

42. A computer-implemented method for identifying SSOs able to modulate expression of a target protein in a cell by promoting incorporation of a pseudoexon into the mature mRNA upon binding to the pre-mRNA in the region +9 to +39 downstream to the 5' splice site of said pseudoexon, the method comprising: a) providing a one or more gene sequences comprising one or more identified pseudoexons; such as in a database format; b) determining for the one or more gene sequences; if the pseudoexon meets the following criteria: Pseudoexon length <160 nt; Pseudoexon length >30 nt; The last 3 nt of the pseudoexon are different from TAG; Donor splice site has a MaxEnt score ≥ 4.33 ; Donor splice site has a MaxEnt score ≤ 10.06 ; and Acceptor splice site has a MaxEnt score ≥ 3.63 ; c) determining for the one or more gene sequences if the sequence region +9 to +39 downstream to the 5' splice site of said pseudoexon meets the following criteria: Total pyrimidines ≤ 20 ; Total thymidine bases ≤ 12 ; Total thymidine bases ≥ 4 ; Total guanine bases ≤ 12 ; Maximum length of thymidine polymer ≤ 4 ; Maximum length of pyrimidine polymer ≤ 10 ; Minimum length of purine polymer ≥ 3 ; and Maximum number of guanine polymers of at least 3 nt length ≤ 2 ; wherein, if one or more gene sequences meet the criteria according to point b) and point c), said region +9 to +39 downstream to the 5' splice site of said pseudoexon is considered a target for an SSO able to, in vivo, hybridize to the pre-mRNA of said gene within the region +9 to +39 downstream to the 5' splice site of said pseudoexon; and resulting in said pseudoexon becoming part of the mature mRNA to a greater extent compared to corresponding pre-mRNA not contacted with the SSO.

43. The computer-implemented method according to claim 42, wherein hybridization of the SSO to the pre-mRNA in vivo results in: decreasing the level of mRNA encoding the functional protein; and/or decreasing expression of the functional protein; and/or loss of function of the functional protein; and/or a new function of the functional protein; and/or mis-localization of the protein; and/or mis-localization of the mRNA encoding the functional protein.
